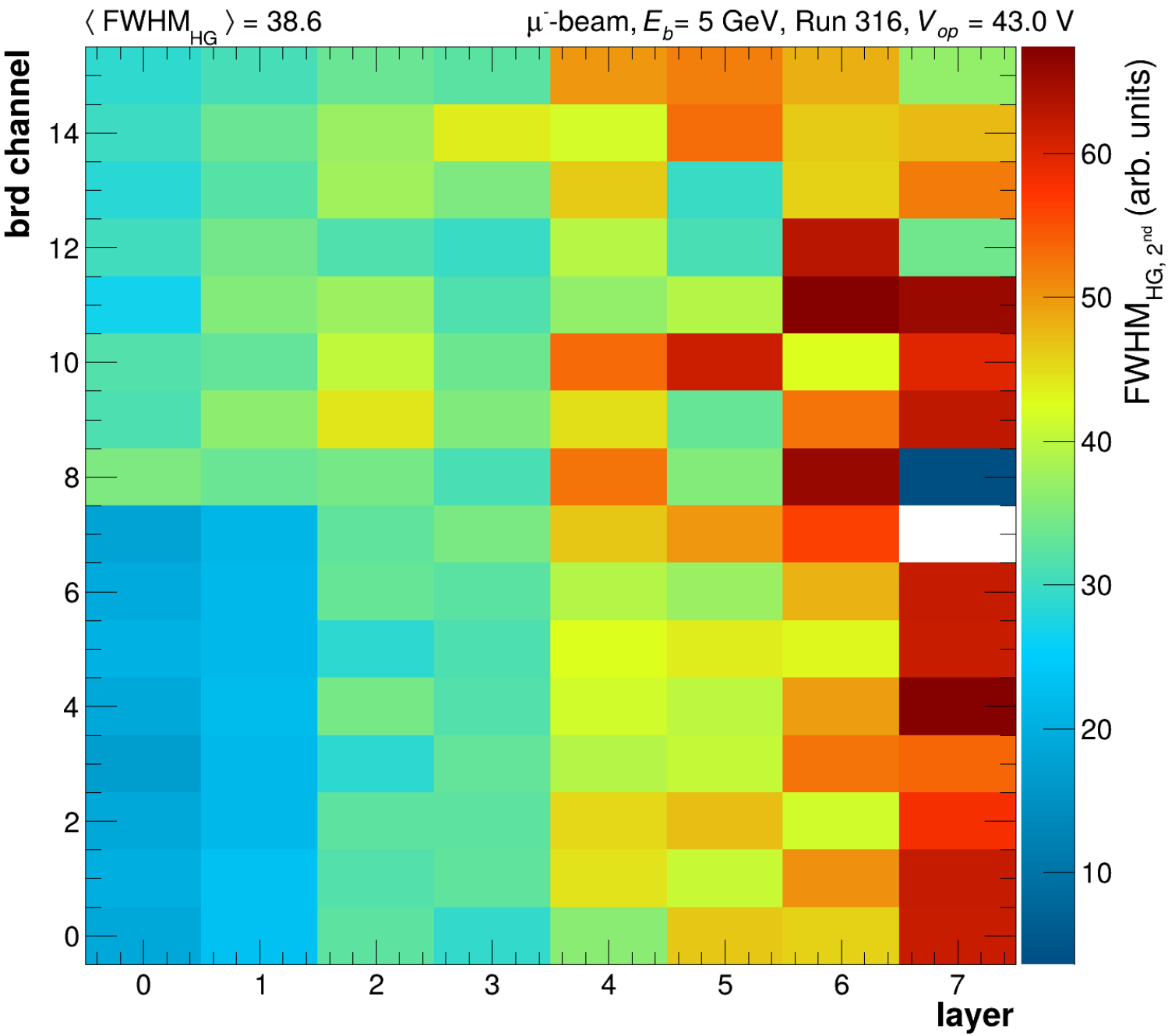
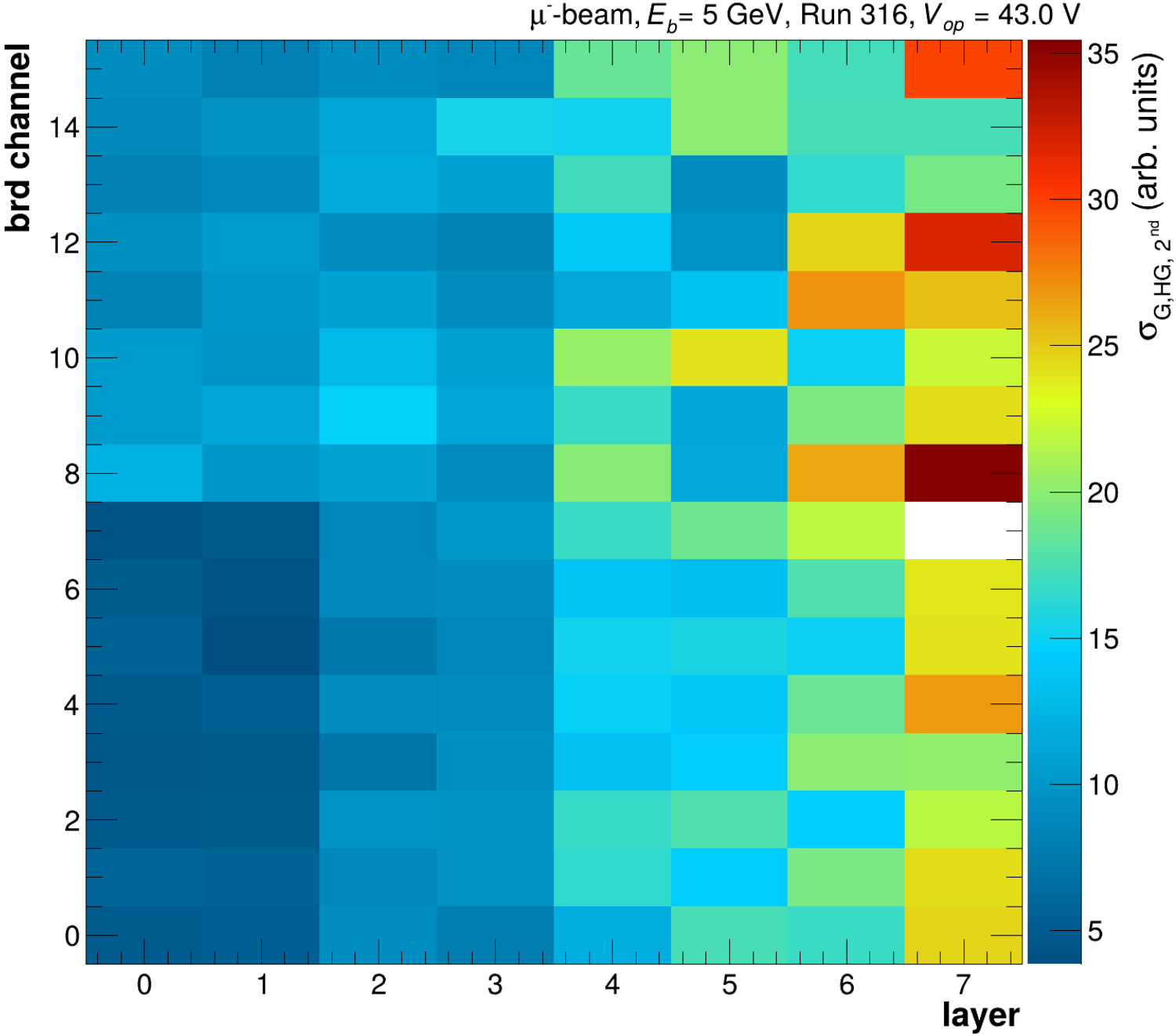
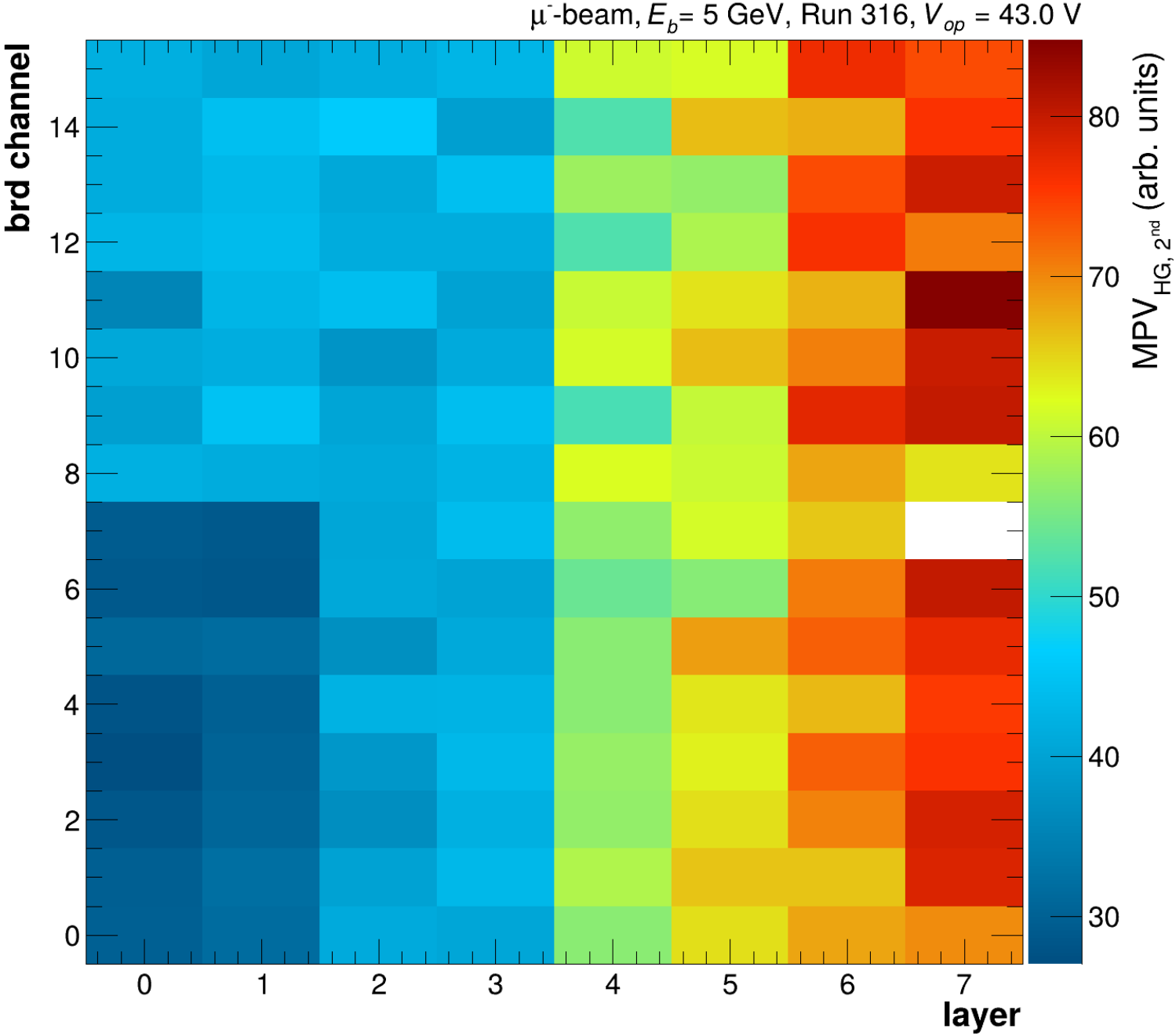
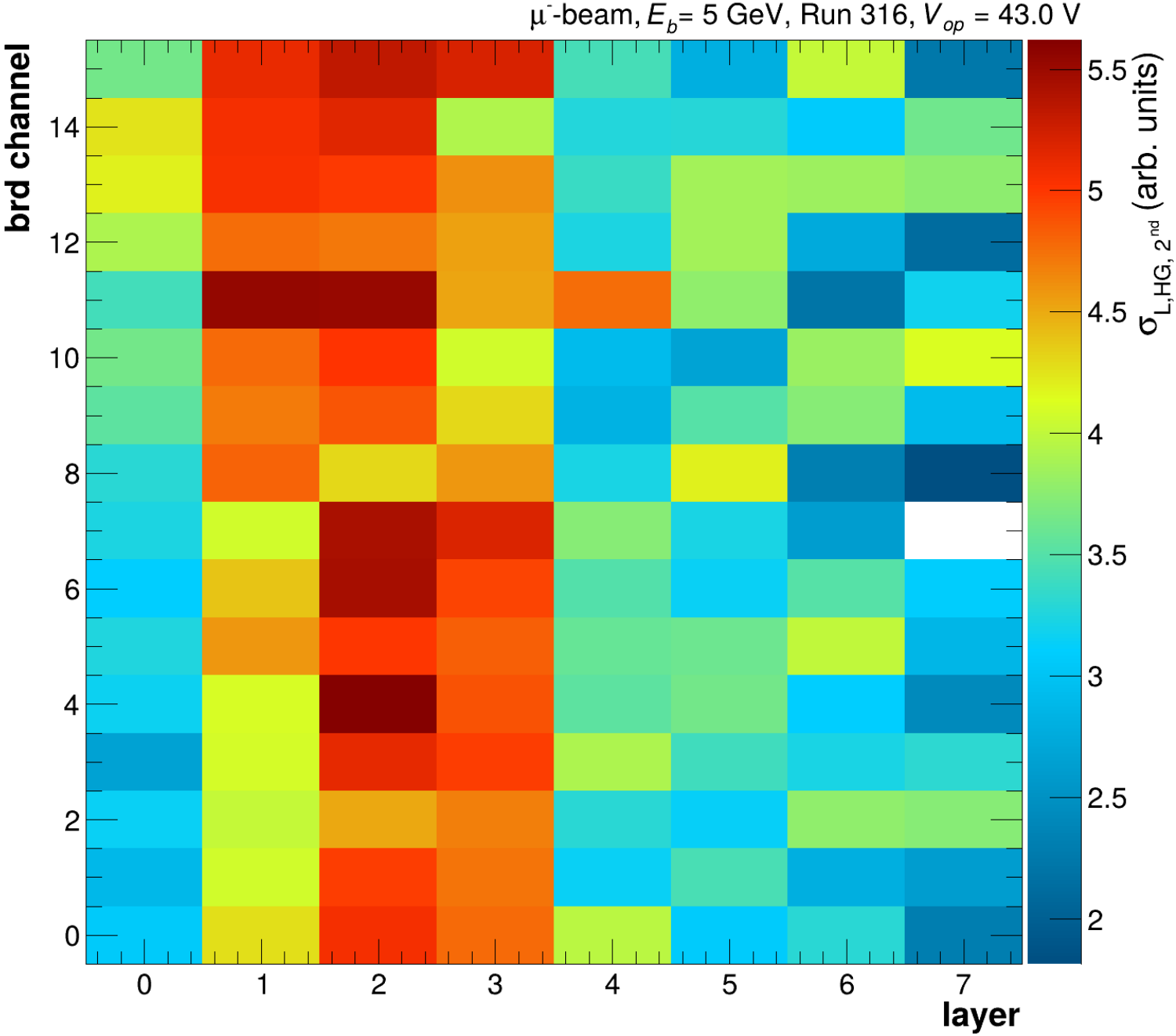
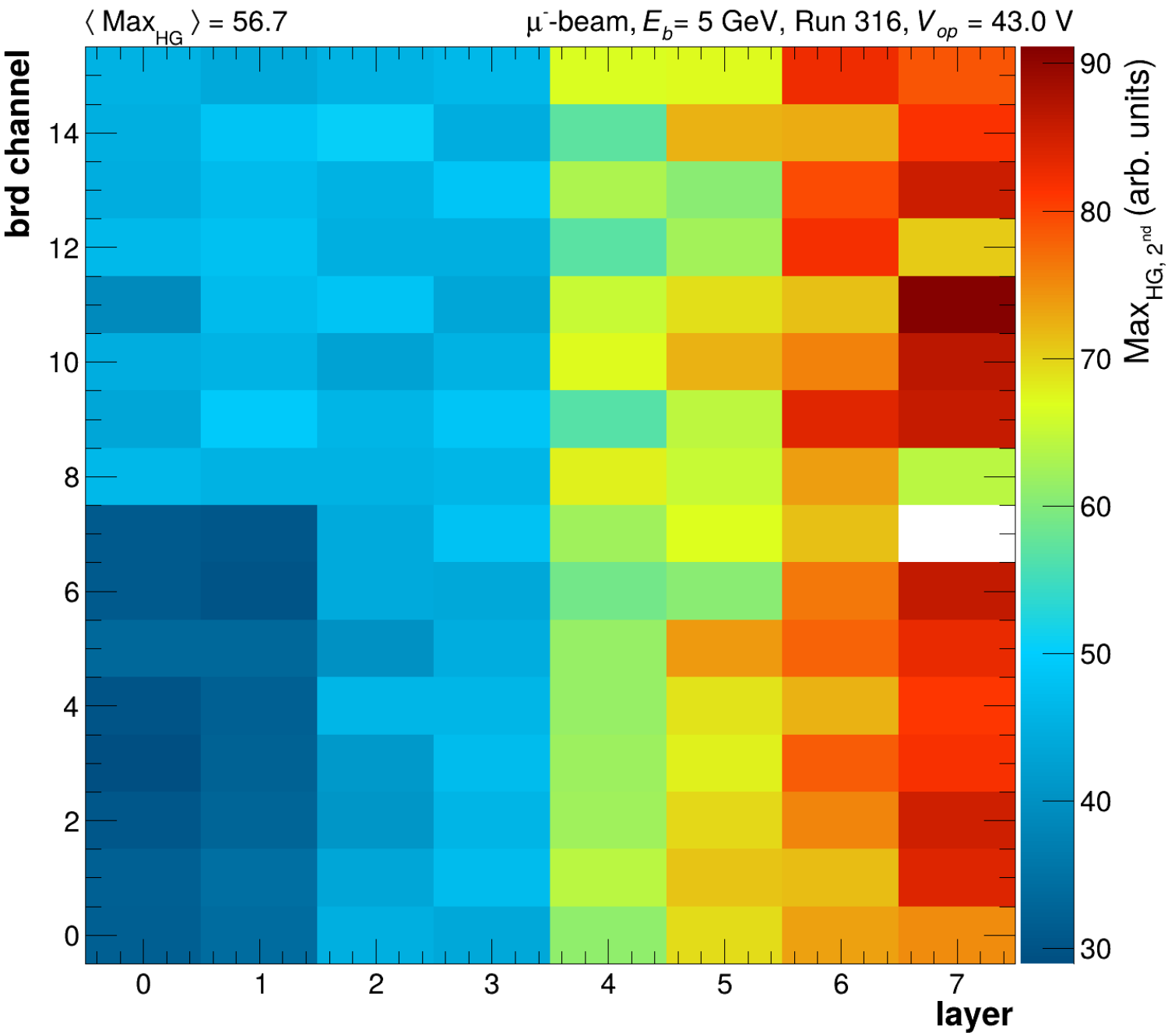
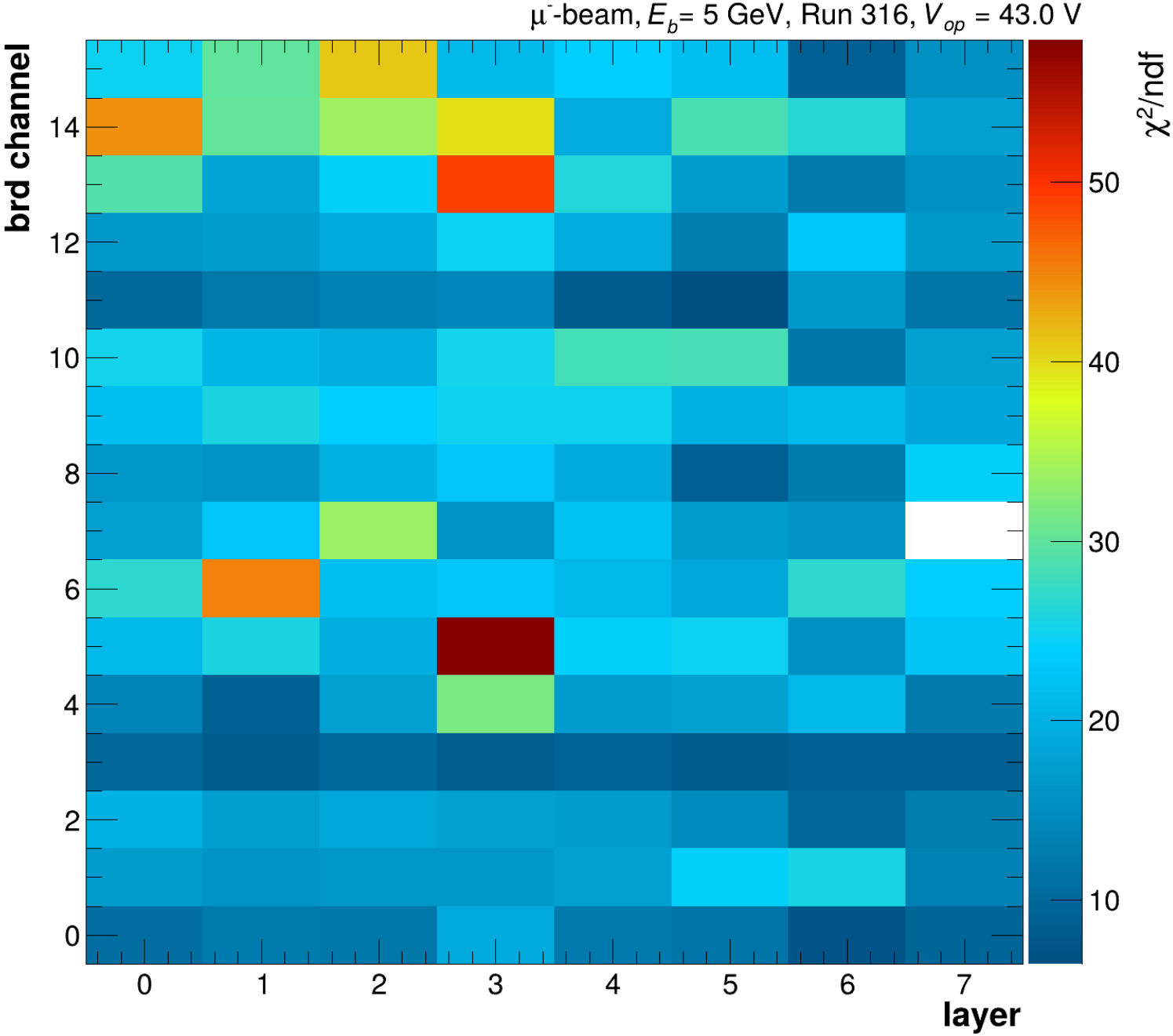

⟨ FWHM_HG ⟩ = 38.6
μ⁻-beam, E_b = 5 GeV, Run 316, V_op = 43.0 V
brd channel
layer
FWHM_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
20
30
40
50
60

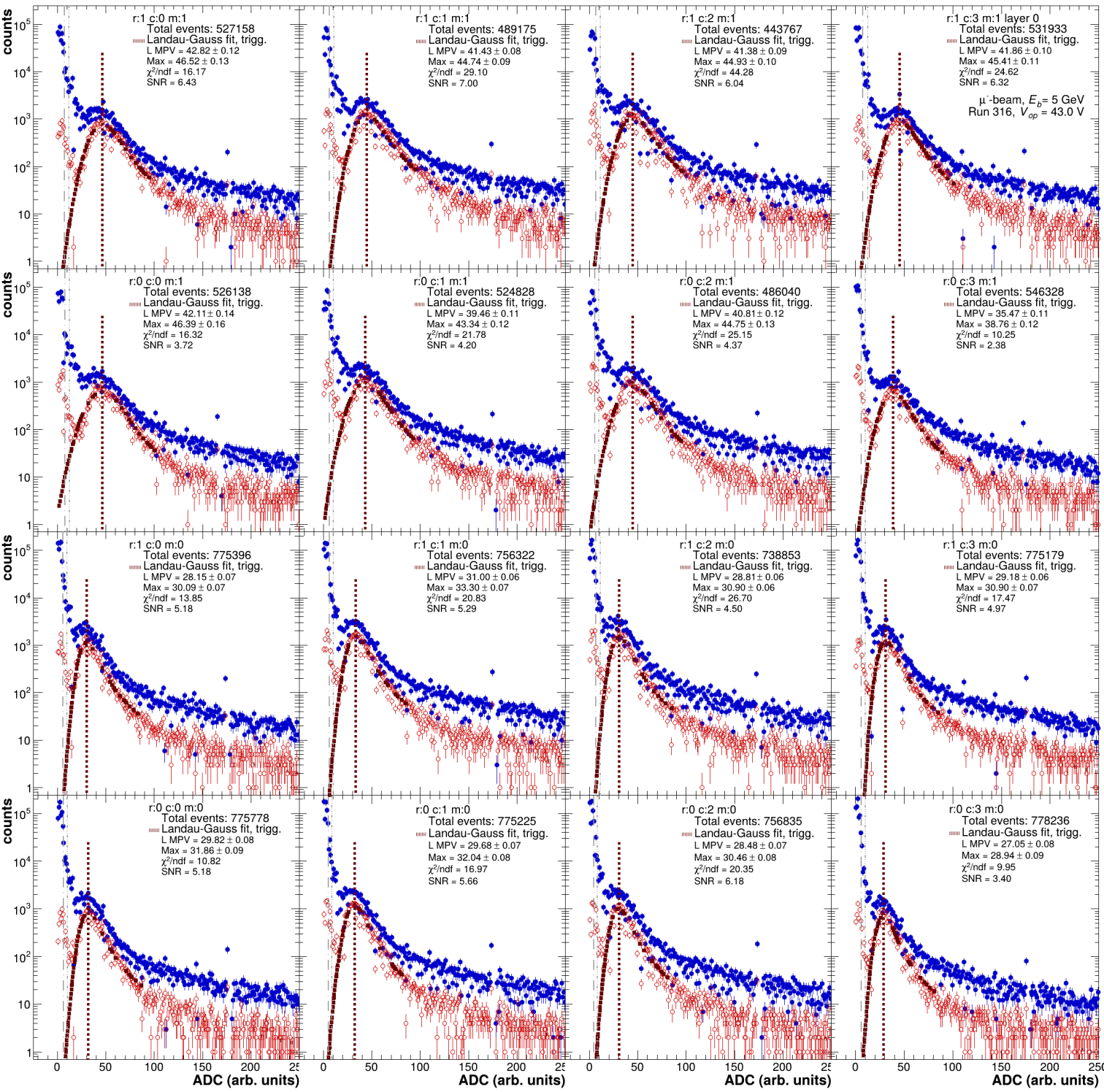

μ⁻-beam, E_b= 5 GeV, Run 316, V_{op} = 43.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
5
10
15
20
25
30
35

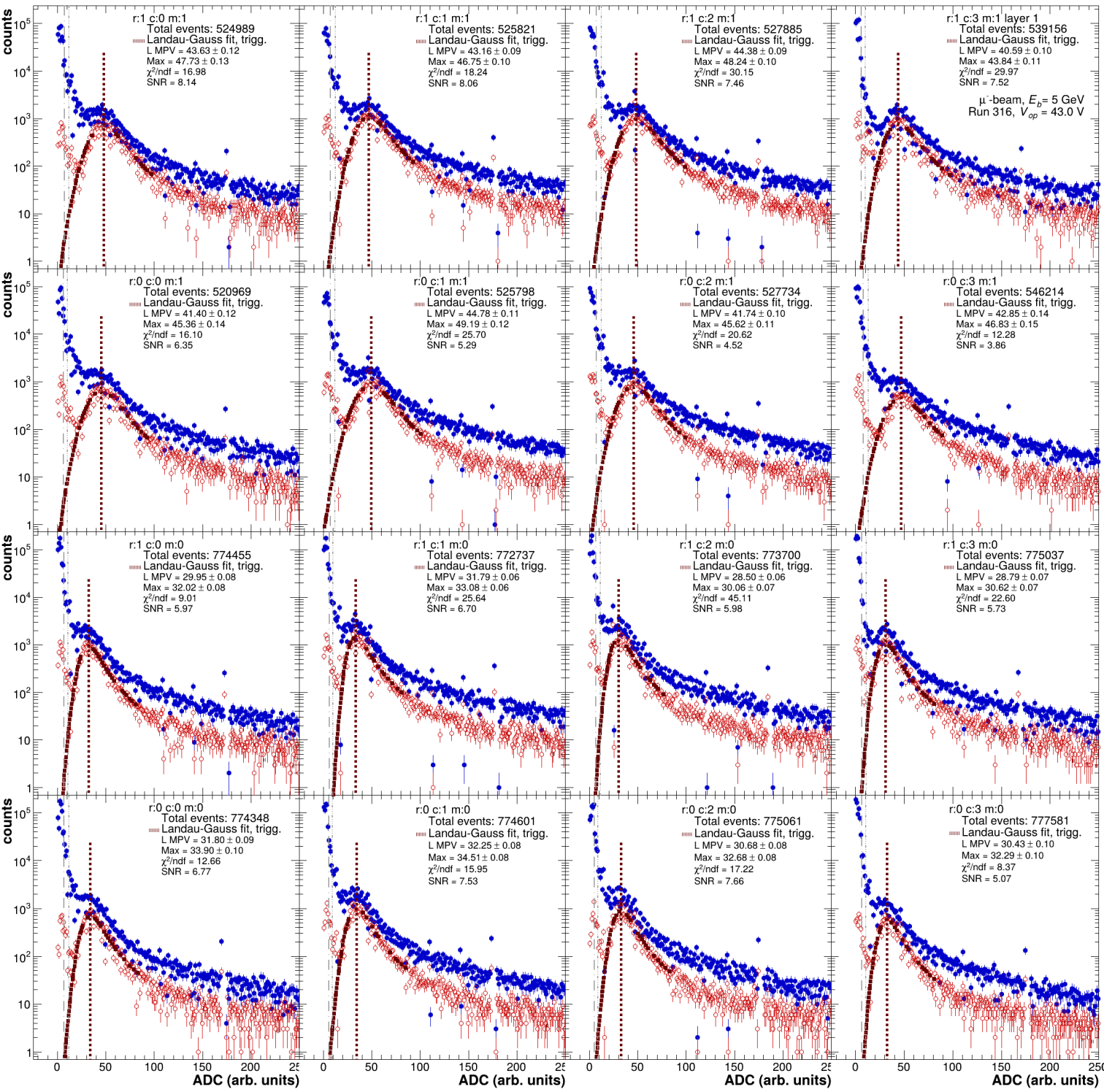

μ⁻-beam, E_b= 5 GeV, Run 316, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
30
40
50
60
70
80
$MPV_{HG,\ 2^{nd}}$ (arb. units)

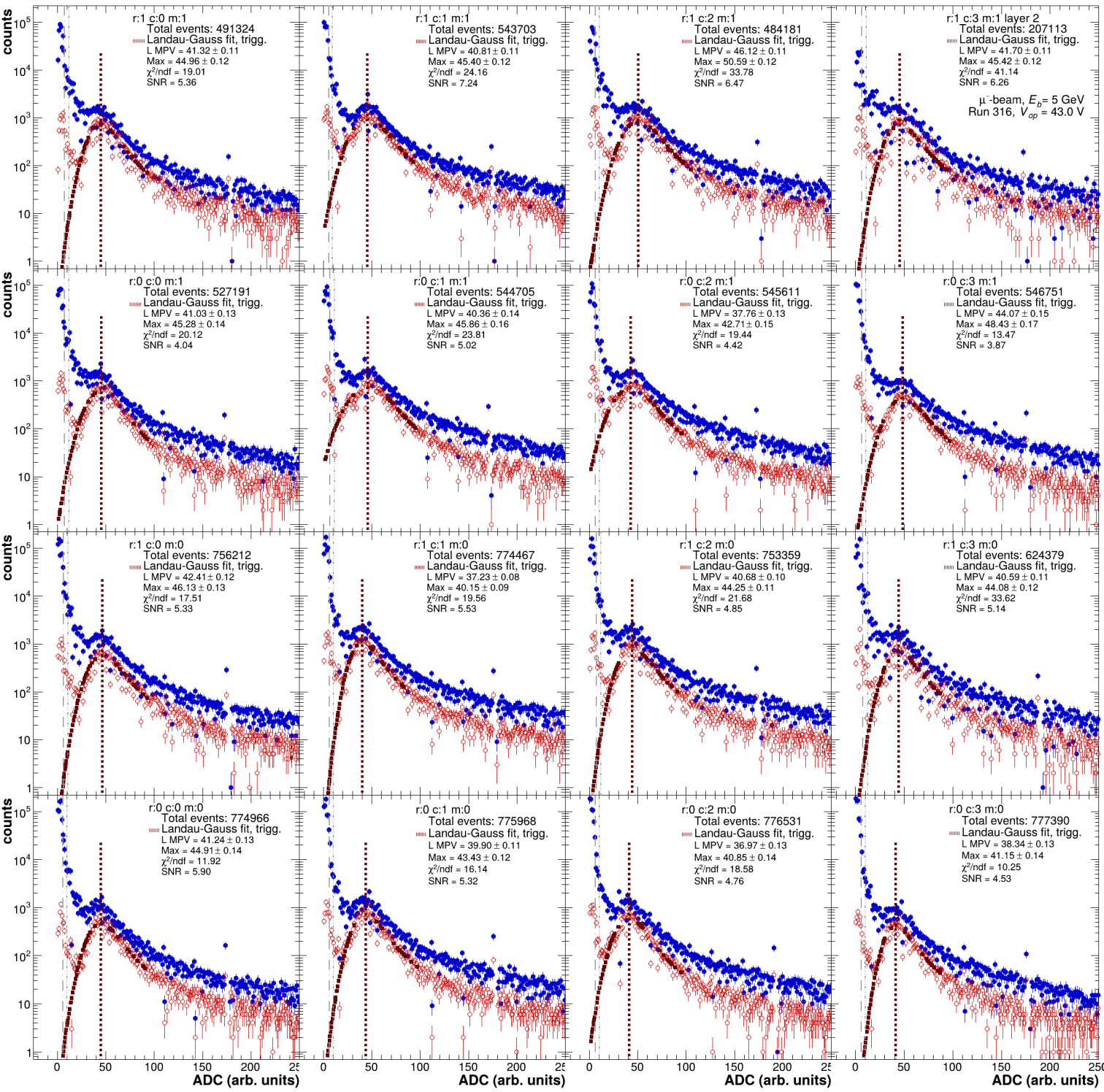

μ⁻-beam, E_b= 5 GeV, Run 316, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
2
2.5
3
3.5
4
4.5
5
5.5

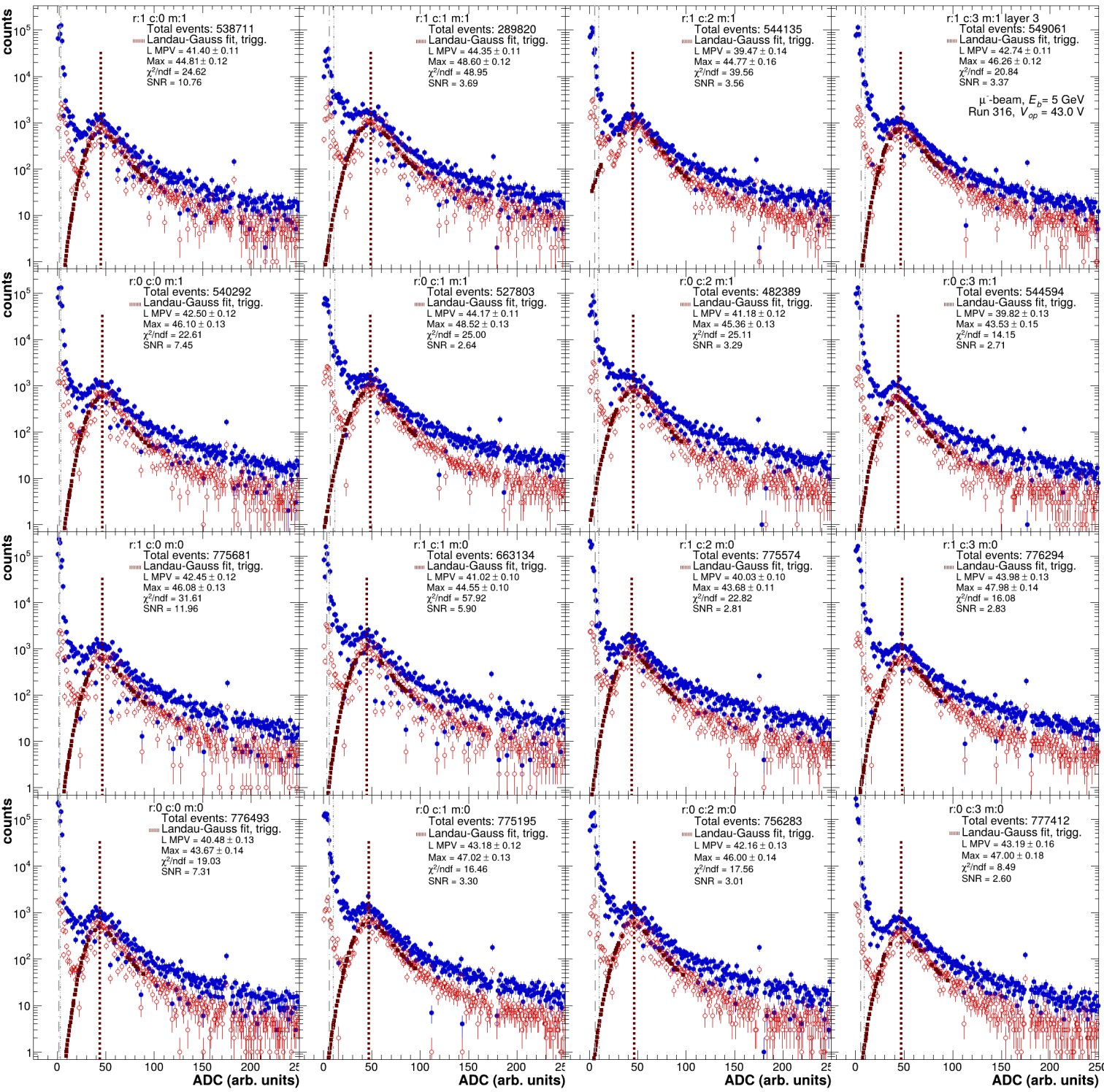

⟨ Max_HG ⟩ = 56.7
μ⁻-beam, E_b = 5 GeV, Run 316, V_op = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
Max_HG, 2nd (arb. units)
30
40
50
60
70
80
90

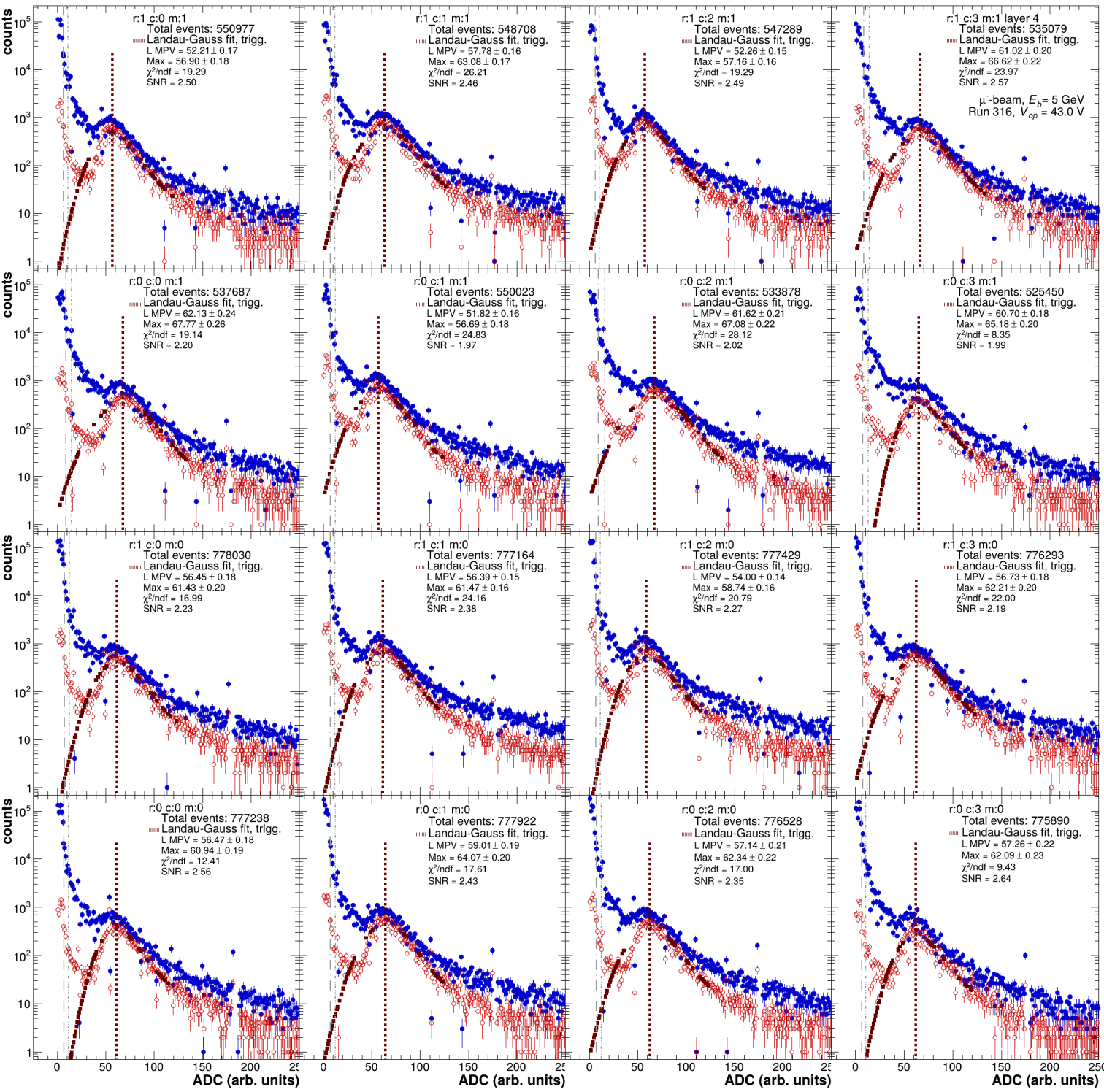

μ⁻-beam, E_b= 5 GeV, Run 316, V_{op} = 43.0 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
20
30
40
50

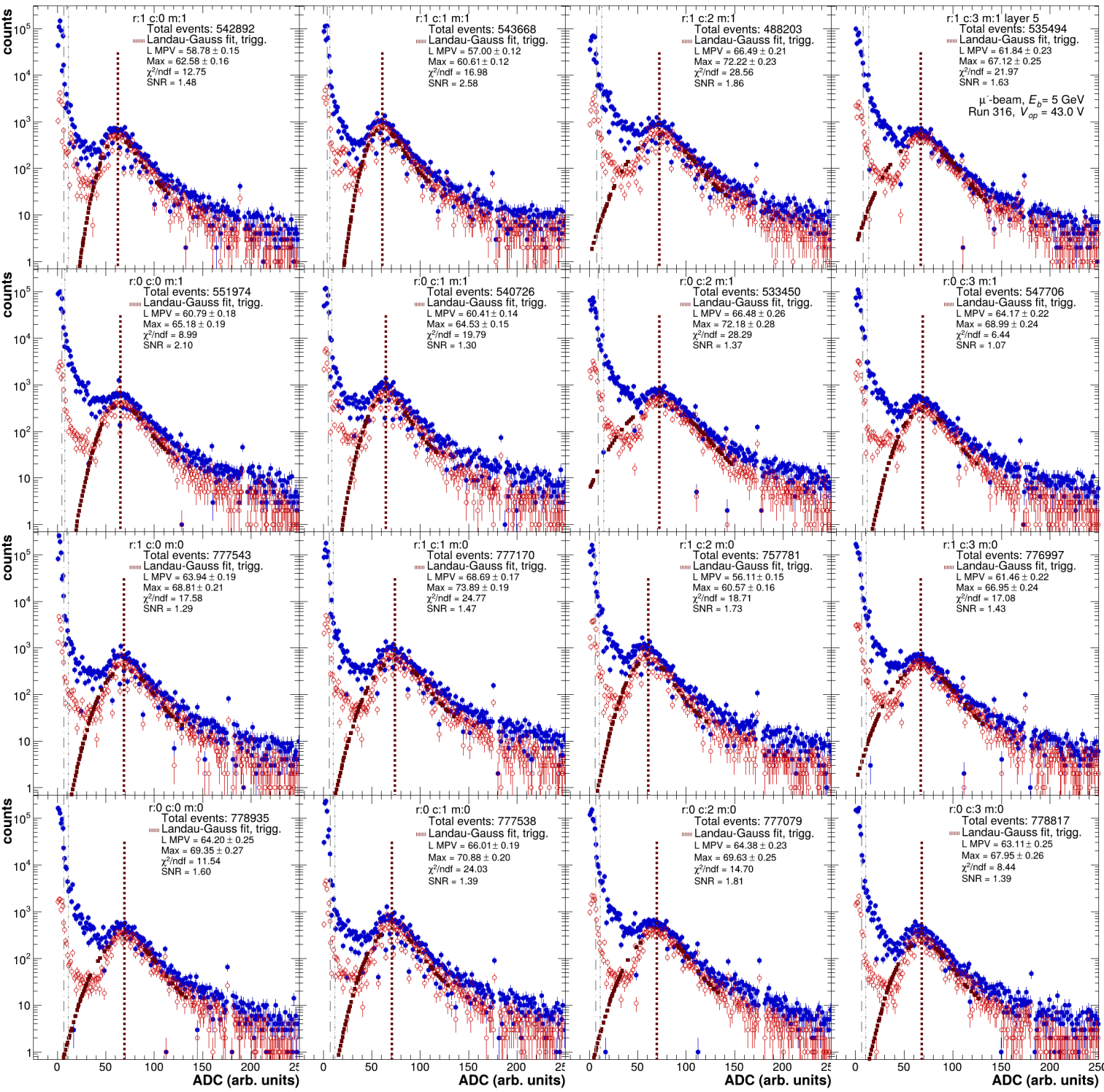

r:1 c:0 m:1
Total events: 527158
Landau-Gauss fit, trigg.
L MPV = 42.82 ± 0.12
Max = 46.52 ± 0.13
χ²/ndf = 16.17
SNR = 6.43
r:1 c:1 m:1
Total events: 489175
Landau-Gauss fit, trigg.
L MPV = 41.43 ± 0.08
Max = 44.74 ± 0.09
χ²/ndf = 29.10
SNR = 7.00
r:1 c:2 m:1
Total events: 443767
Landau-Gauss fit, trigg.
L MPV = 41.38 ± 0.09
Max = 44.93 ± 0.10
χ²/ndf = 44.28
SNR = 6.04
r:1 c:3 m:1 layer 0
Total events: 531933
Landau-Gauss fit, trigg.
L MPV = 41.86 ± 0.10
Max = 45.41 ± 0.11
χ²/ndf = 24.62
SNR = 6.32
μ⁻-beam, E_b = 5 GeV
Run 316, V_op = 43.0 V
r:0 c:0 m:1
Total events: 526138
Landau-Gauss fit, trigg.
L MPV = 42.11 ± 0.14
Max = 46.39 ± 0.16
χ²/ndf = 16.32
SNR = 3.72
r:0 c:1 m:1
Total events: 524828
Landau-Gauss fit, trigg.
L MPV = 39.46 ± 0.11
Max = 43.34 ± 0.12
χ²/ndf = 21.78
SNR = 4.20
r:0 c:2 m:1
Total events: 486040
Landau-Gauss fit, trigg.
L MPV = 40.81 ± 0.12
Max = 44.75 ± 0.13
χ²/ndf = 25.15
SNR = 4.37
r:0 c:3 m:1
Total events: 546328
Landau-Gauss fit, trigg.
L MPV = 35.47 ± 0.11
Max = 38.76 ± 0.12
χ²/ndf = 10.25
SNR = 2.38
r:1 c:0 m:0
Total events: 775396
Landau-Gauss fit, trigg.
L MPV = 28.15 ± 0.07
Max = 30.09 ± 0.07
χ²/ndf = 13.85
SNR = 5.18
r:1 c:1 m:0
Total events: 756322
Landau-Gauss fit, trigg.
L MPV = 31.00 ± 0.06
Max = 33.30 ± 0.07
χ²/ndf = 20.83
SNR = 5.29
r:1 c:2 m:0
Total events: 738853
Landau-Gauss fit, trigg.
L MPV = 28.81 ± 0.06
Max = 30.90 ± 0.06
χ²/ndf = 26.70
SNR = 4.50
r:1 c:3 m:0
Total events: 775179
Landau-Gauss fit, trigg.
L MPV = 29.18 ± 0.06
Max = 30.90 ± 0.07
χ²/ndf = 17.47
SNR = 4.97
r:0 c:0 m:0
Total events: 775778
Landau-Gauss fit, trigg.
L MPV = 29.82 ± 0.08
Max = 31.86 ± 0.09
χ²/ndf = 10.82
SNR = 5.18
r:0 c:1 m:0
Total events: 775225
Landau-Gauss fit, trigg.
L MPV = 29.68 ± 0.07
Max = 32.04 ± 0.08
χ²/ndf = 16.97
SNR = 5.66
r:0 c:2 m:0
Total events: 756835
Landau-Gauss fit, trigg.
L MPV = 28.48 ± 0.07
Max = 30.46 ± 0.08
χ²/ndf = 20.35
SNR = 6.18
r:0 c:3 m:0
Total events: 778236
Landau-Gauss fit, trigg.
L MPV = 27.05 ± 0.08
Max = 28.94 ± 0.09
χ²/ndf = 9.95
SNR = 3.40
counts
ADC (arb. units)

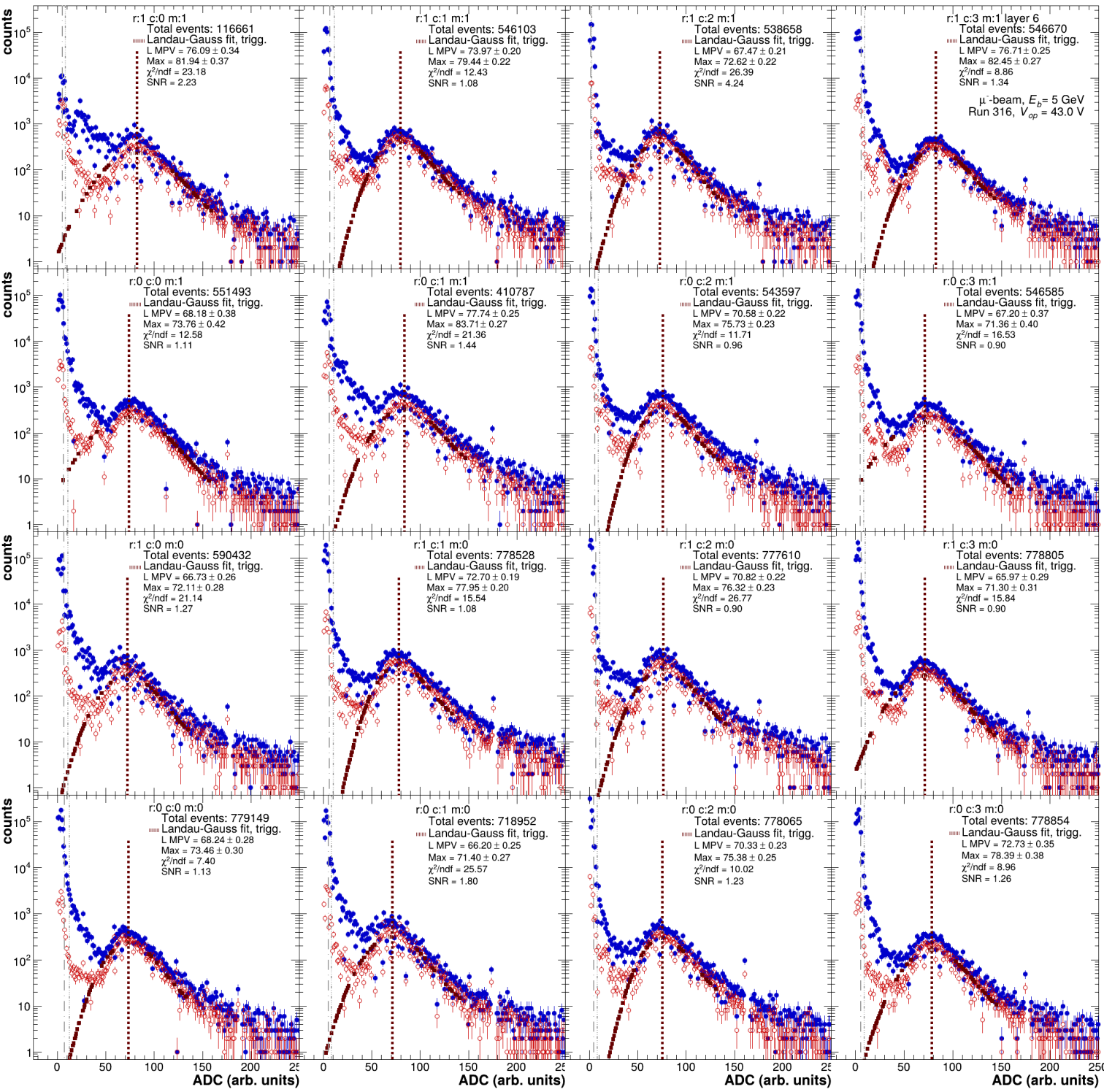

r:1 c:0 m:1
Total events: 524989
Landau-Gauss fit, trigg.
L MPV = 43.63 ± 0.12
Max = 47.73 ± 0.13
χ²/ndf = 16.98
SNR = 8.14
r:1 c:1 m:1
Total events: 525821
Landau-Gauss fit, trigg.
L MPV = 43.16 ± 0.09
Max = 46.75 ± 0.10
χ²/ndf = 18.24
SNR = 8.06
r:1 c:2 m:1
Total events: 527885
Landau-Gauss fit, trigg.
L MPV = 44.38 ± 0.09
Max = 48.24 ± 0.10
χ²/ndf = 30.15
SNR = 7.46
r:1 c:3 m:1 layer 1
Total events: 539156
Landau-Gauss fit, trigg.
L MPV = 40.59 ± 0.10
Max = 43.84 ± 0.11
χ²/ndf = 29.97
SNR = 7.52
μ⁻-beam, E_b = 5 GeV
Run 316, V_op = 43.0 V
r:0 c:0 m:1
Total events: 520969
Landau-Gauss fit, trigg.
L MPV = 41.40 ± 0.12
Max = 45.36 ± 0.14
χ²/ndf = 16.10
SNR = 6.35
r:0 c:1 m:1
Total events: 525798
Landau-Gauss fit, trigg.
L MPV = 44.78 ± 0.11
Max = 49.19 ± 0.12
χ²/ndf = 25.70
SNR = 5.29
r:0 c:2 m:1
Total events: 527734
Landau-Gauss fit, trigg.
L MPV = 41.74 ± 0.10
Max = 45.62 ± 0.11
χ²/ndf = 20.62
SNR = 4.52
r:0 c:3 m:1
Total events: 546214
Landau-Gauss fit, trigg.
L MPV = 42.85 ± 0.14
Max = 46.83 ± 0.15
χ²/ndf = 12.28
SNR = 3.86
r:1 c:0 m:0
Total events: 774455
Landau-Gauss fit, trigg.
L MPV = 29.95 ± 0.08
Max = 32.02 ± 0.08
χ²/ndf = 9.01
SNR = 5.97
r:1 c:1 m:0
Total events: 772737
Landau-Gauss fit, trigg.
L MPV = 31.79 ± 0.06
Max = 33.08 ± 0.06
χ²/ndf = 25.64
SNR = 6.70
r:1 c:2 m:0
Total events: 773700
Landau-Gauss fit, trigg.
L MPV = 28.50 ± 0.06
Max = 30.06 ± 0.07
χ²/ndf = 45.11
SNR = 5.98
r:1 c:3 m:0
Total events: 775037
Landau-Gauss fit, trigg.
L MPV = 28.79 ± 0.07
Max = 30.62 ± 0.07
χ²/ndf = 22.60
SNR = 5.73
r:0 c:0 m:0
Total events: 774348
Landau-Gauss fit, trigg.
L MPV = 31.80 ± 0.09
Max = 33.90 ± 0.10
χ²/ndf = 12.66
SNR = 6.77
r:0 c:1 m:0
Total events: 774601
Landau-Gauss fit, trigg.
L MPV = 32.25 ± 0.08
Max = 34.51 ± 0.08
χ²/ndf = 15.95
SNR = 7.53
r:0 c:2 m:0
Total events: 775061
Landau-Gauss fit, trigg.
L MPV = 30.68 ± 0.08
Max = 32.68 ± 0.08
χ²/ndf = 17.22
SNR = 7.66
r:0 c:3 m:0
Total events: 777581
Landau-Gauss fit, trigg.
L MPV = 30.43 ± 0.10
Max = 32.29 ± 0.10
χ²/ndf = 8.37
SNR = 5.07
counts
ADC (arb. units)

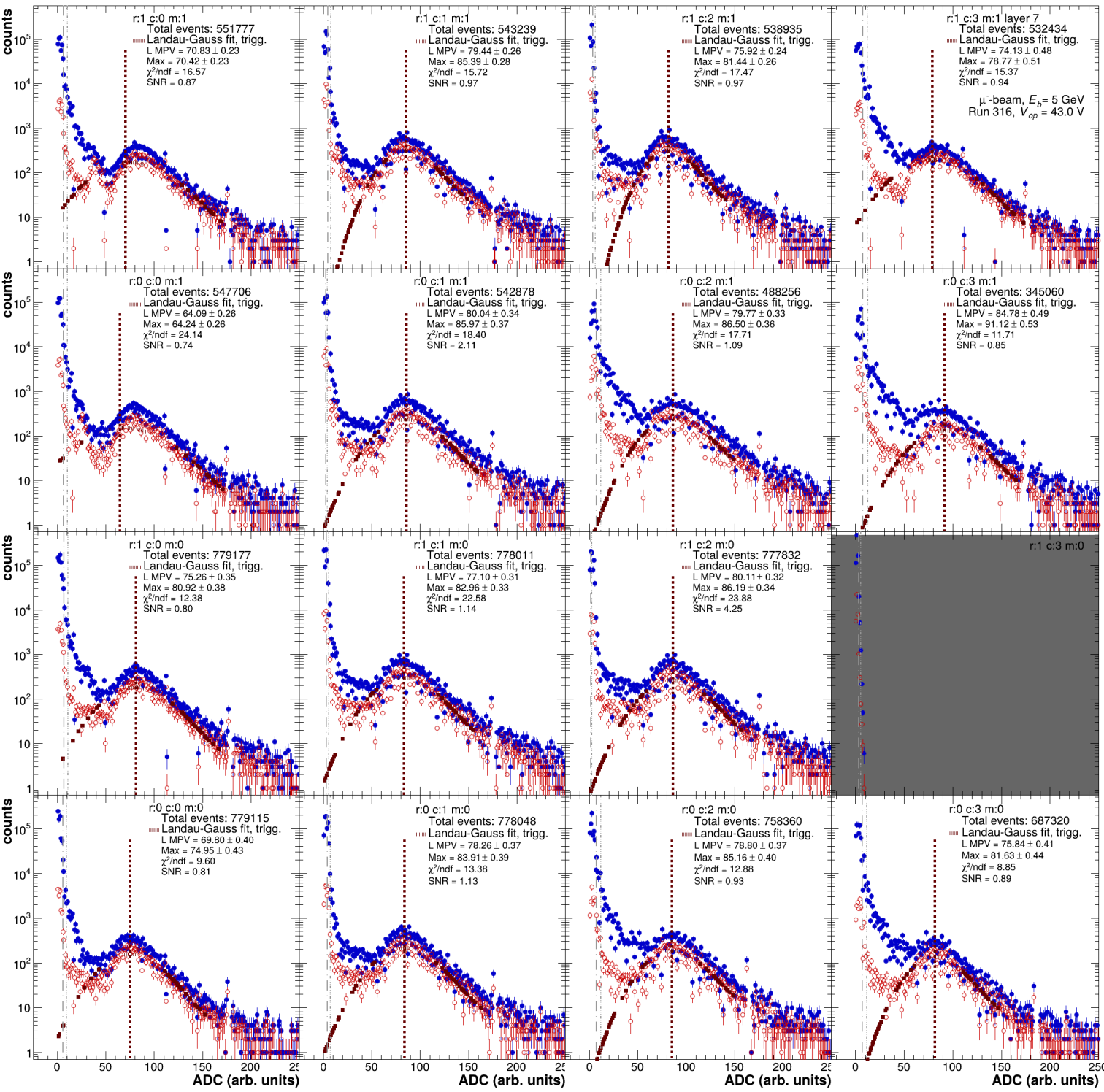

r:1 c:0 m:1
Total events: 491324
Landau-Gauss fit, trigg.
L MPV = 41.32 ± 0.11
Max = 44.96 ± 0.12
χ²/ndf = 19.01
SNR = 5.36
r:1 c:1 m:1
Total events: 543703
Landau-Gauss fit, trigg.
L MPV = 40.81 ± 0.11
Max = 45.40 ± 0.12
χ²/ndf = 24.16
SNR = 7.24
r:1 c:2 m:1
Total events: 484181
Landau-Gauss fit, trigg.
L MPV = 46.12 ± 0.11
Max = 50.59 ± 0.12
χ²/ndf = 33.78
SNR = 6.47
r:1 c:3 m:1 layer 2
Total events: 207113
Landau-Gauss fit, trigg.
L MPV = 41.70 ± 0.11
Max = 45.42 ± 0.12
χ²/ndf = 41.14
SNR = 6.26
μ⁻-beam, E_b = 5 GeV
Run 316, V_op = 43.0 V
r:0 c:0 m:1
Total events: 527191
Landau-Gauss fit, trigg.
L MPV = 41.03 ± 0.13
Max = 45.28 ± 0.14
χ²/ndf = 20.12
SNR = 4.04
r:0 c:1 m:1
Total events: 544705
Landau-Gauss fit, trigg.
L MPV = 40.36 ± 0.14
Max = 45.86 ± 0.16
χ²/ndf = 23.81
SNR = 5.02
r:0 c:2 m:1
Total events: 545611
Landau-Gauss fit, trigg.
L MPV = 37.76 ± 0.13
Max = 42.71 ± 0.15
χ²/ndf = 19.44
SNR = 4.42
r:0 c:3 m:1
Total events: 546751
Landau-Gauss fit, trigg.
L MPV = 44.07 ± 0.15
Max = 48.43 ± 0.17
χ²/ndf = 13.47
SNR = 3.87
r:1 c:0 m:0
Total events: 756212
Landau-Gauss fit, trigg.
L MPV = 42.41 ± 0.12
Max = 46.13 ± 0.13
χ²/ndf = 17.51
SNR = 5.33
r:1 c:1 m:0
Total events: 774467
Landau-Gauss fit, trigg.
L MPV = 37.23 ± 0.08
Max = 40.15 ± 0.09
χ²/ndf = 19.56
SNR = 5.53
r:1 c:2 m:0
Total events: 753359
Landau-Gauss fit, trigg.
L MPV = 40.68 ± 0.10
Max = 44.25 ± 0.11
χ²/ndf = 21.68
SNR = 4.85
r:1 c:3 m:0
Total events: 624379
Landau-Gauss fit, trigg.
L MPV = 40.59 ± 0.11
Max = 44.08 ± 0.12
χ²/ndf = 33.62
SNR = 5.14
r:0 c:0 m:0
Total events: 774966
Landau-Gauss fit, trigg.
L MPV = 41.24 ± 0.13
Max = 44.91 ± 0.14
χ²/ndf = 11.92
SNR = 5.90
r:0 c:1 m:0
Total events: 775968
Landau-Gauss fit, trigg.
L MPV = 39.90 ± 0.11
Max = 43.43 ± 0.12
χ²/ndf = 16.14
SNR = 5.32
r:0 c:2 m:0
Total events: 776531
Landau-Gauss fit, trigg.
L MPV = 36.97 ± 0.13
Max = 40.85 ± 0.14
χ²/ndf = 18.58
SNR = 4.76
r:0 c:3 m:0
Total events: 777390
Landau-Gauss fit, trigg.
L MPV = 38.34 ± 0.13
Max = 41.15 ± 0.14
χ²/ndf = 10.25
SNR = 4.53
counts
ADC (arb. units)

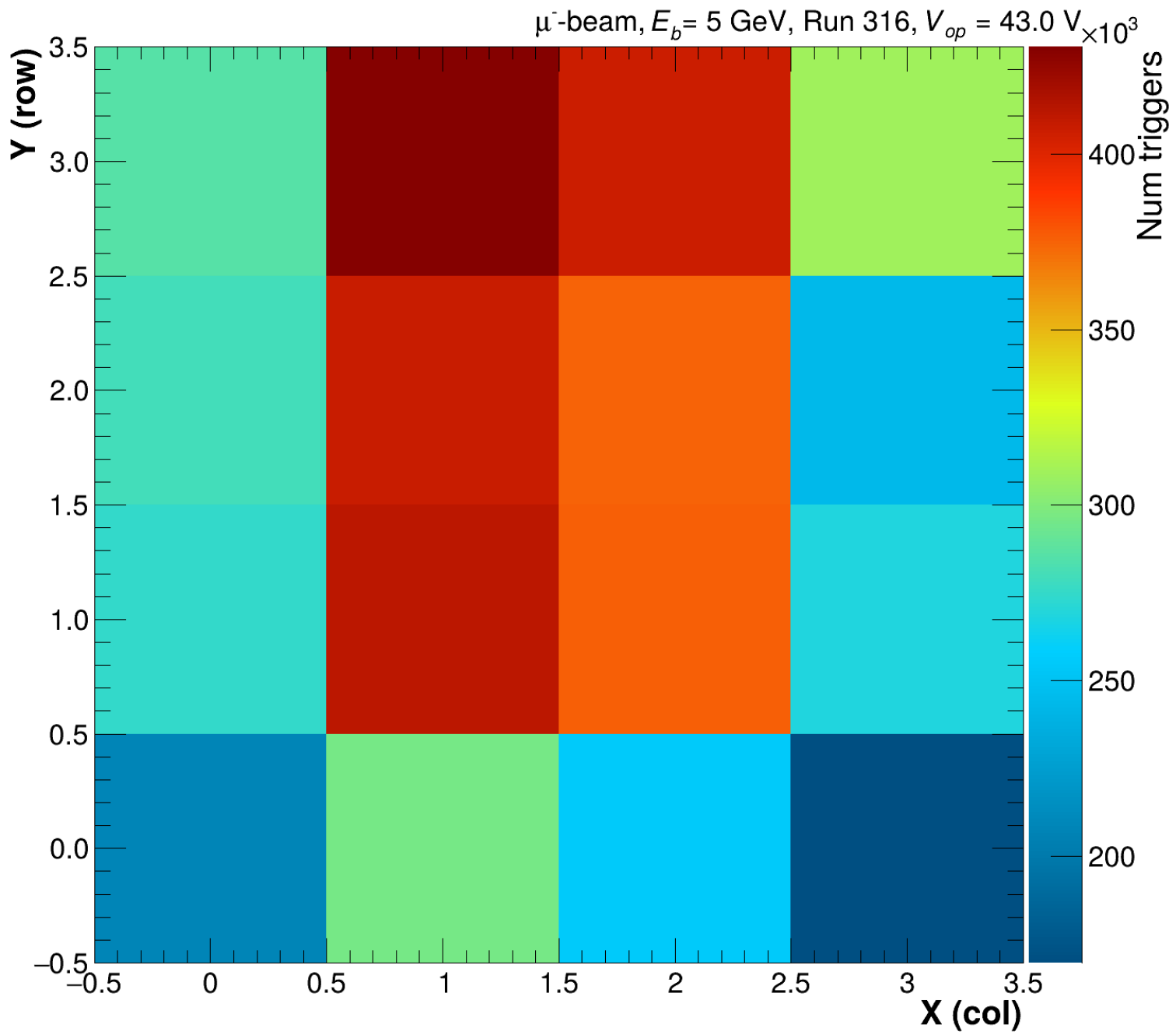

r:1 c:0 m:1
Total events: 538711
Landau-Gauss fit, trigg.
L MPV = 41.40 ± 0.11
Max = 44.81 ± 0.12
χ²/ndf = 24.62
SNR = 10.76
r:1 c:1 m:1
Total events: 289820
Landau-Gauss fit, trigg.
L MPV = 44.35 ± 0.11
Max = 48.60 ± 0.12
χ²/ndf = 48.95
SNR = 3.69
r:1 c:2 m:1
Total events: 544135
Landau-Gauss fit, trigg.
L MPV = 39.47 ± 0.14
Max = 44.77 ± 0.16
χ²/ndf = 39.56
SNR = 3.56
r:1 c:3 m:1 layer 3
Total events: 549061
Landau-Gauss fit, trigg.
L MPV = 42.74 ± 0.11
Max = 46.26 ± 0.12
χ²/ndf = 20.84
SNR = 3.37
μ⁻-beam, E_b = 5 GeV
Run 316, V_op = 43.0 V
r:0 c:0 m:1
Total events: 540292
Landau-Gauss fit, trigg.
L MPV = 42.50 ± 0.12
Max = 46.10 ± 0.13
χ²/ndf = 22.61
SNR = 7.45
r:0 c:1 m:1
Total events: 527803
Landau-Gauss fit, trigg.
L MPV = 44.17 ± 0.11
Max = 48.52 ± 0.13
χ²/ndf = 25.00
SNR = 2.64
r:0 c:2 m:1
Total events: 482389
Landau-Gauss fit, trigg.
L MPV = 41.18 ± 0.12
Max = 45.36 ± 0.13
χ²/ndf = 25.11
SNR = 3.29
r:0 c:3 m:1
Total events: 544594
Landau-Gauss fit, trigg.
L MPV = 39.82 ± 0.13
Max = 43.53 ± 0.15
χ²/ndf = 14.15
SNR = 2.71
r:1 c:0 m:0
Total events: 775681
Landau-Gauss fit, trigg.
L MPV = 42.45 ± 0.12
Max = 46.08 ± 0.13
χ²/ndf = 31.61
SNR = 11.96
r:1 c:1 m:0
Total events: 663134
Landau-Gauss fit, trigg.
L MPV = 41.02 ± 0.10
Max = 44.55 ± 0.10
χ²/ndf = 57.92
SNR = 5.90
r:1 c:2 m:0
Total events: 775574
Landau-Gauss fit, trigg.
L MPV = 40.03 ± 0.10
Max = 43.68 ± 0.11
χ²/ndf = 22.82
SNR = 2.81
r:1 c:3 m:0
Total events: 776294
Landau-Gauss fit, trigg.
L MPV = 43.98 ± 0.13
Max = 47.98 ± 0.14
χ²/ndf = 16.08
SNR = 2.83
r:0 c:0 m:0
Total events: 776493
Landau-Gauss fit, trigg.
L MPV = 40.48 ± 0.13
Max = 43.67 ± 0.14
χ²/ndf = 19.03
SNR = 7.31
r:0 c:1 m:0
Total events: 775195
Landau-Gauss fit, trigg.
L MPV = 43.18 ± 0.12
Max = 47.02 ± 0.13
χ²/ndf = 16.46
SNR = 3.30
r:0 c:2 m:0
Total events: 756283
Landau-Gauss fit, trigg.
L MPV = 42.16 ± 0.13
Max = 46.00 ± 0.14
χ²/ndf = 17.56
SNR = 3.01
r:0 c:3 m:0
Total events: 777412
Landau-Gauss fit, trigg.
L MPV = 43.19 ± 0.16
Max = 47.00 ± 0.18
χ²/ndf = 8.49
SNR = 2.60
counts
ADC (arb. units)

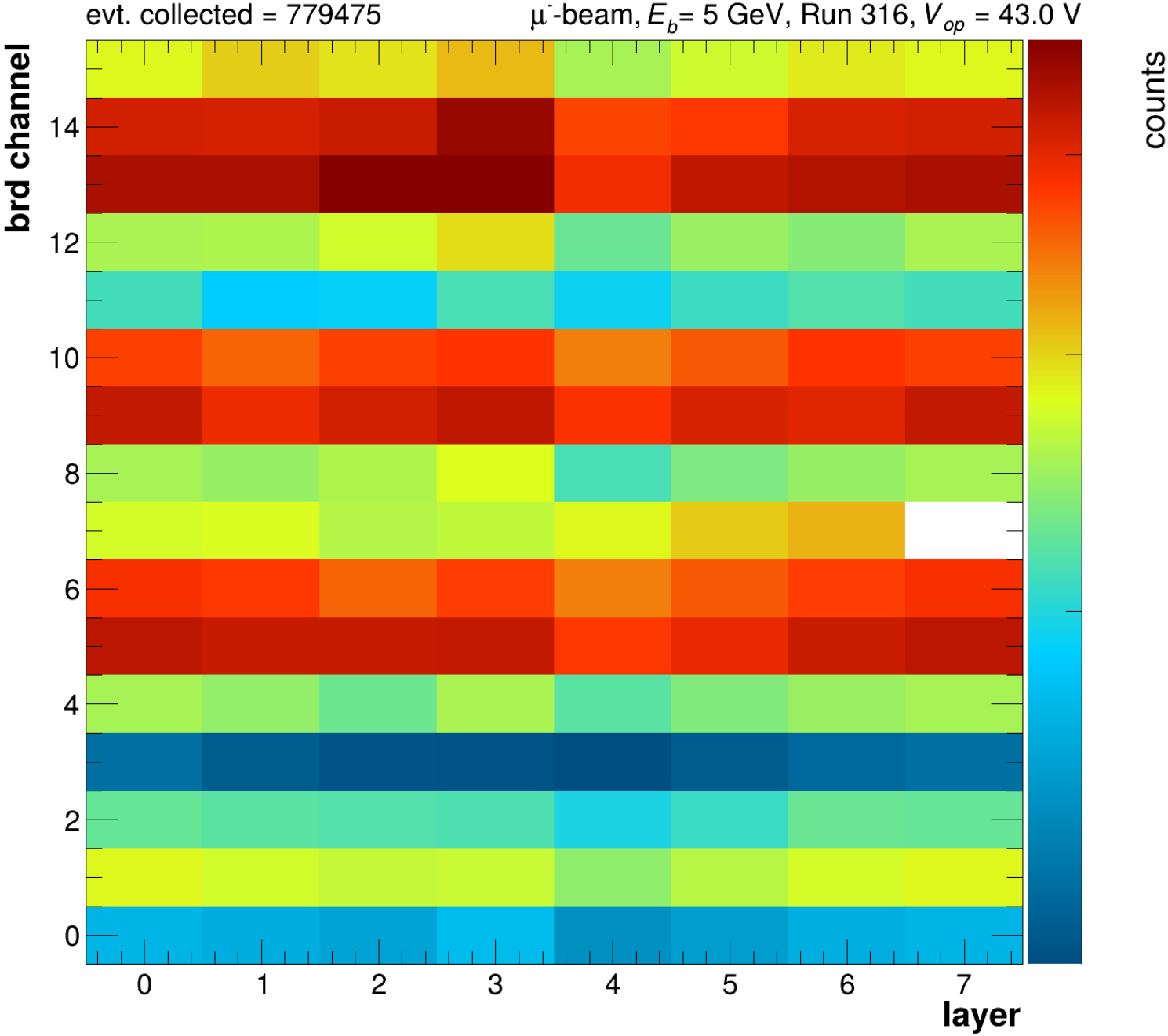

r:1 c:0 m:1
Total events: 550977
Landau-Gauss fit, trigg.
L MPV = 52.21 ± 0.17
Max = 56.90 ± 0.18
χ²/ndf = 19.29
SNR = 2.50
r:1 c:1 m:1
Total events: 548708
Landau-Gauss fit, trigg.
L MPV = 57.78 ± 0.16
Max = 63.08 ± 0.17
χ²/ndf = 26.21
SNR = 2.46
r:1 c:2 m:1
Total events: 547289
Landau-Gauss fit, trigg.
L MPV = 52.26 ± 0.15
Max = 57.16 ± 0.16
χ²/ndf = 19.29
SNR = 2.49
r:1 c:3 m:1 layer 4
Total events: 535079
Landau-Gauss fit, trigg.
L MPV = 61.02 ± 0.20
Max = 66.62 ± 0.22
χ²/ndf = 23.97
SNR = 2.57
μ⁻-beam, E_b = 5 GeV
Run 316, V_op = 43.0 V
r:0 c:0 m:1
Total events: 537687
Landau-Gauss fit, trigg.
L MPV = 62.13 ± 0.24
Max = 67.77 ± 0.26
χ²/ndf = 19.14
SNR = 2.20
r:0 c:1 m:1
Total events: 550023
Landau-Gauss fit, trigg.
L MPV = 51.82 ± 0.16
Max = 56.69 ± 0.18
χ²/ndf = 24.83
SNR = 1.97
r:0 c:2 m:1
Total events: 533878
Landau-Gauss fit, trigg.
L MPV = 61.62 ± 0.21
Max = 67.08 ± 0.22
χ²/ndf = 28.12
SNR = 2.02
r:0 c:3 m:1
Total events: 525450
Landau-Gauss fit, trigg.
L MPV = 60.70 ± 0.18
Max = 65.18 ± 0.20
χ²/ndf = 8.35
SNR = 1.99
r:1 c:0 m:0
Total events: 778030
Landau-Gauss fit, trigg.
L MPV = 56.45 ± 0.18
Max = 61.43 ± 0.20
χ²/ndf = 16.99
SNR = 2.23
r:1 c:1 m:0
Total events: 777164
Landau-Gauss fit, trigg.
L MPV = 56.39 ± 0.15
Max = 61.47 ± 0.16
χ²/ndf = 24.16
SNR = 2.38
r:1 c:2 m:0
Total events: 777429
Landau-Gauss fit, trigg.
L MPV = 54.00 ± 0.14
Max = 58.74 ± 0.16
χ²/ndf = 20.79
SNR = 2.27
r:1 c:3 m:0
Total events: 776293
Landau-Gauss fit, trigg.
L MPV = 56.73 ± 0.18
Max = 62.21 ± 0.20
χ²/ndf = 22.00
SNR = 2.19
r:0 c:0 m:0
Total events: 777238
Landau-Gauss fit, trigg.
L MPV = 56.47 ± 0.18
Max = 60.94 ± 0.19
χ²/ndf = 12.41
SNR = 2.56
r:0 c:1 m:0
Total events: 777922
Landau-Gauss fit, trigg.
L MPV = 59.01 ± 0.19
Max = 64.07 ± 0.20
χ²/ndf = 17.61
SNR = 2.43
r:0 c:2 m:0
Total events: 776528
Landau-Gauss fit, trigg.
L MPV = 57.14 ± 0.21
Max = 62.34 ± 0.22
χ²/ndf = 17.00
SNR = 2.35
r:0 c:3 m:0
Total events: 775890
Landau-Gauss fit, trigg.
L MPV = 57.26 ± 0.22
Max = 62.09 ± 0.23
χ²/ndf = 9.43
SNR = 2.64
counts
ADC (arb. units)

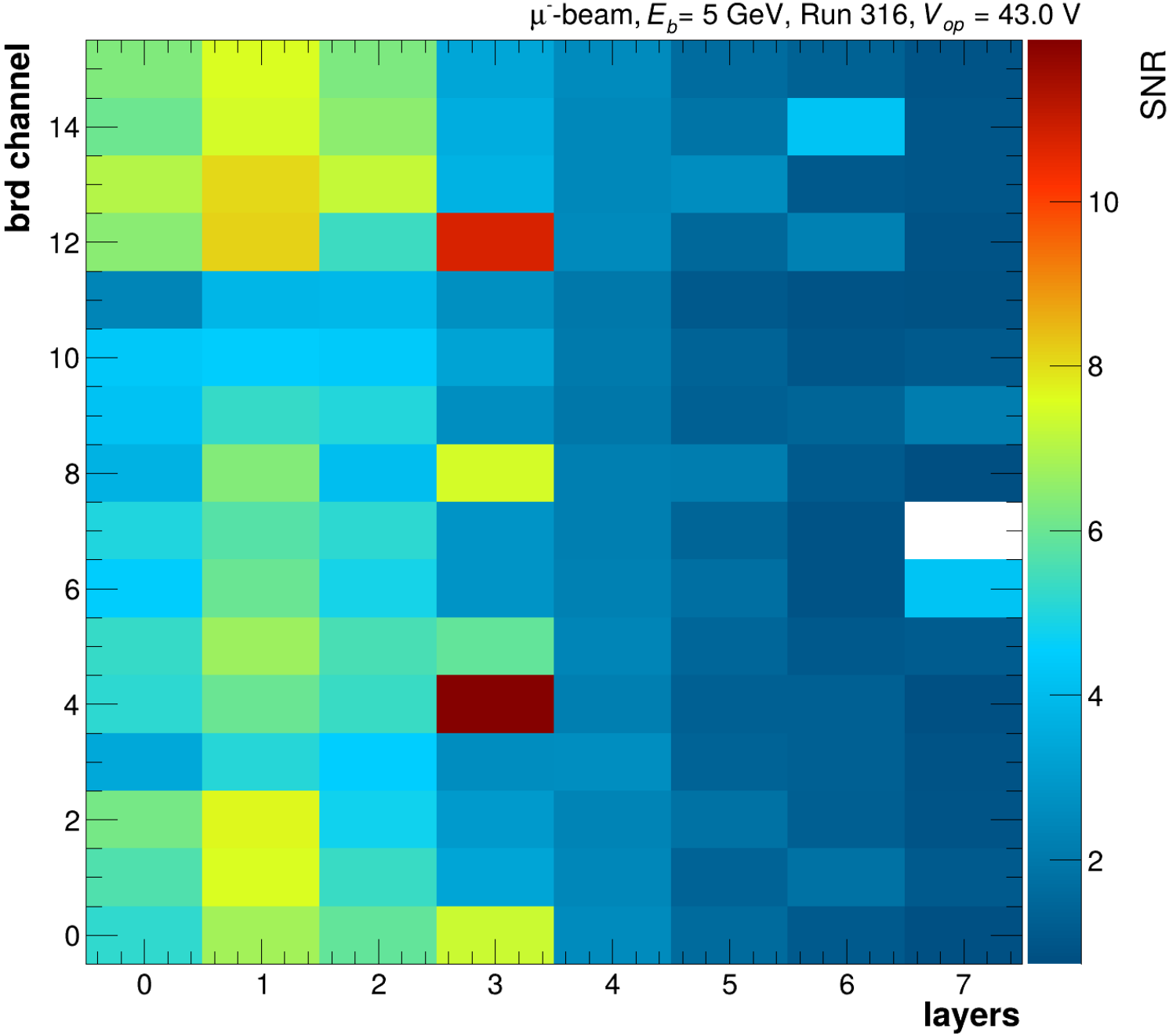

r:1 c:0 m:1
Total events: 542892
Landau-Gauss fit, trigg.
L MPV = 58.78 ± 0.15
Max = 62.58 ± 0.16
χ²/ndf = 12.75
SNR = 1.48
r:1 c:1 m:1
Total events: 543668
Landau-Gauss fit, trigg.
L MPV = 57.00 ± 0.12
Max = 60.61 ± 0.12
χ²/ndf = 16.98
SNR = 2.58
r:1 c:2 m:1
Total events: 488203
Landau-Gauss fit, trigg.
L MPV = 66.49 ± 0.21
Max = 72.22 ± 0.23
χ²/ndf = 28.56
SNR = 1.86
r:1 c:3 m:1 layer 5
Total events: 535494
Landau-Gauss fit, trigg.
L MPV = 61.84 ± 0.23
Max = 67.12 ± 0.25
χ²/ndf = 21.97
SNR = 1.63
μ⁻-beam, E_b = 5 GeV
Run 316, V_op = 43.0 V
r:0 c:0 m:1
Total events: 551974
Landau-Gauss fit, trigg.
L MPV = 60.79 ± 0.18
Max = 65.18 ± 0.19
χ²/ndf = 8.99
SNR = 2.10
r:0 c:1 m:1
Total events: 540726
Landau-Gauss fit, trigg.
L MPV = 60.41 ± 0.14
Max = 64.53 ± 0.15
χ²/ndf = 19.79
SNR = 1.30
r:0 c:2 m:1
Total events: 533450
Landau-Gauss fit, trigg.
L MPV = 66.48 ± 0.26
Max = 72.18 ± 0.28
χ²/ndf = 28.29
SNR = 1.37
r:0 c:3 m:1
Total events: 547706
Landau-Gauss fit, trigg.
L MPV = 64.17 ± 0.22
Max = 68.99 ± 0.24
χ²/ndf = 6.44
SNR = 1.07
r:1 c:0 m:0
Total events: 777543
Landau-Gauss fit, trigg.
L MPV = 63.94 ± 0.19
Max = 68.81 ± 0.21
χ²/ndf = 17.58
SNR = 1.29
r:1 c:1 m:0
Total events: 777170
Landau-Gauss fit, trigg.
L MPV = 68.69 ± 0.17
Max = 73.89 ± 0.19
χ²/ndf = 24.77
SNR = 1.47
r:1 c:2 m:0
Total events: 757781
Landau-Gauss fit, trigg.
L MPV = 56.11 ± 0.15
Max = 60.57 ± 0.16
χ²/ndf = 18.71
SNR = 1.73
r:1 c:3 m:0
Total events: 776997
Landau-Gauss fit, trigg.
L MPV = 61.46 ± 0.22
Max = 66.95 ± 0.24
χ²/ndf = 17.08
SNR = 1.43
r:0 c:0 m:0
Total events: 778935
Landau-Gauss fit, trigg.
L MPV = 64.20 ± 0.25
Max = 69.35 ± 0.27
χ²/ndf = 11.54
SNR = 1.60
r:0 c:1 m:0
Total events: 777538
Landau-Gauss fit, trigg.
L MPV = 66.01 ± 0.19
Max = 70.88 ± 0.20
χ²/ndf = 24.03
SNR = 1.39
r:0 c:2 m:0
Total events: 777079
Landau-Gauss fit, trigg.
L MPV = 64.38 ± 0.23
Max = 69.63 ± 0.25
χ²/ndf = 14.70
SNR = 1.81
r:0 c:3 m:0
Total events: 778817
Landau-Gauss fit, trigg.
L MPV = 63.11 ± 0.25
Max = 67.95 ± 0.26
χ²/ndf = 8.44
SNR = 1.39
counts
ADC (arb. units)

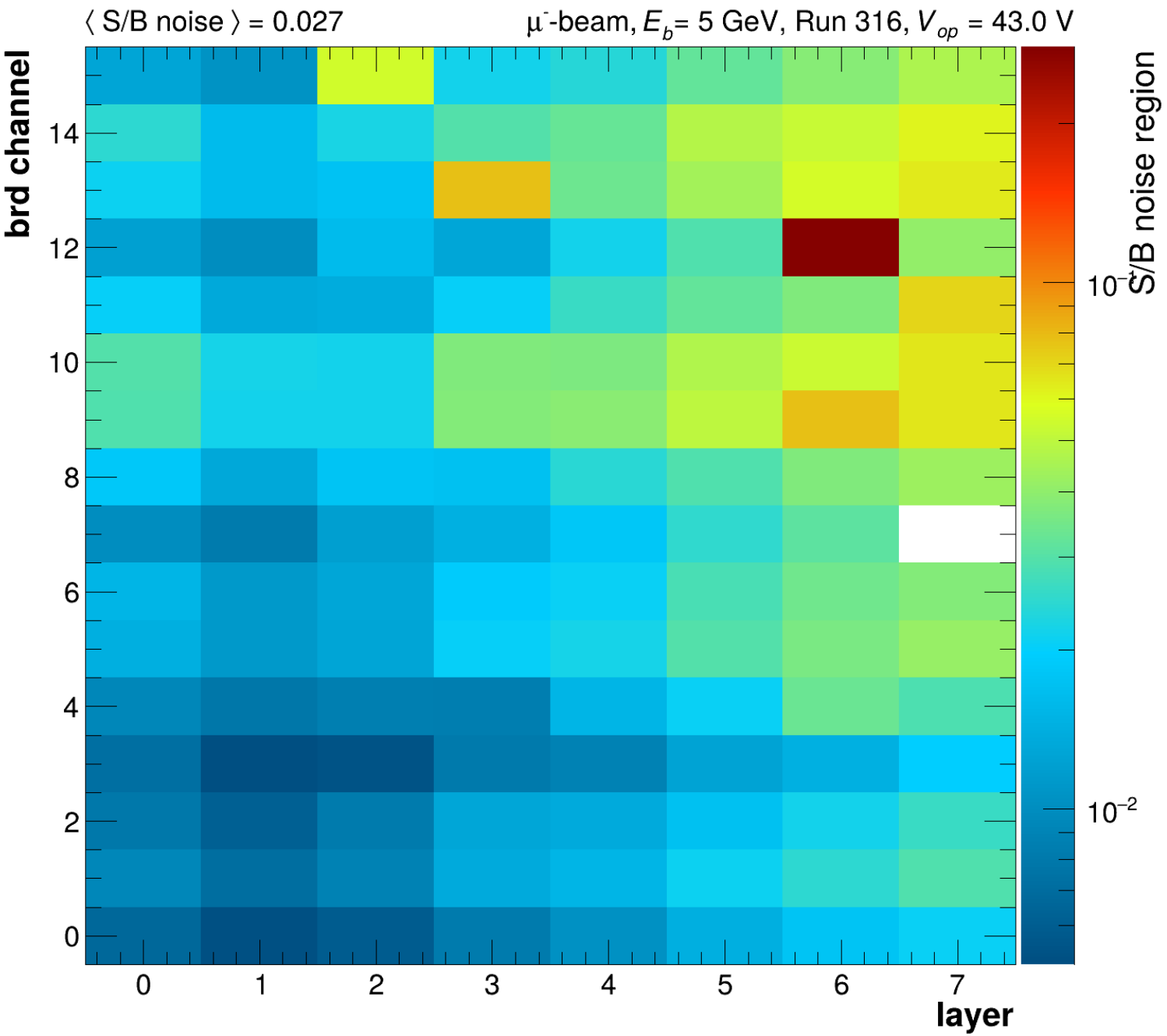

r:1 c:0 m:1
Total events: 116661
Landau-Gauss fit, trigg.
L MPV = 76.09 ± 0.34
Max = 81.94 ± 0.37
χ²/ndf = 23.18
SNR = 2.23
r:1 c:1 m:1
Total events: 546103
Landau-Gauss fit, trigg.
L MPV = 73.97 ± 0.20
Max = 79.44 ± 0.22
χ²/ndf = 12.43
SNR = 1.08
r:1 c:2 m:1
Total events: 538658
Landau-Gauss fit, trigg.
L MPV = 67.47 ± 0.21
Max = 72.62 ± 0.22
χ²/ndf = 26.39
SNR = 4.24
r:1 c:3 m:1 layer 6
Total events: 546670
Landau-Gauss fit, trigg.
L MPV = 76.71 ± 0.25
Max = 82.45 ± 0.27
χ²/ndf = 8.86
SNR = 1.34
μ⁻-beam, E_b = 5 GeV
Run 316, V_op = 43.0 V
r:0 c:0 m:1
Total events: 551493
Landau-Gauss fit, trigg.
L MPV = 68.18 ± 0.38
Max = 73.76 ± 0.42
χ²/ndf = 12.58
SNR = 1.11
r:0 c:1 m:1
Total events: 410787
Landau-Gauss fit, trigg.
L MPV = 77.74 ± 0.25
Max = 83.71 ± 0.27
χ²/ndf = 21.36
SNR = 1.44
r:0 c:2 m:1
Total events: 543597
Landau-Gauss fit, trigg.
L MPV = 70.58 ± 0.22
Max = 75.73 ± 0.23
χ²/ndf = 11.71
SNR = 0.96
r:0 c:3 m:1
Total events: 546585
Landau-Gauss fit, trigg.
L MPV = 67.20 ± 0.37
Max = 71.36 ± 0.40
χ²/ndf = 16.53
SNR = 0.90
r:1 c:0 m:0
Total events: 590432
Landau-Gauss fit, trigg.
L MPV = 66.73 ± 0.26
Max = 72.11 ± 0.28
χ²/ndf = 21.14
SNR = 1.27
r:1 c:1 m:0
Total events: 778528
Landau-Gauss fit, trigg.
L MPV = 72.70 ± 0.19
Max = 77.95 ± 0.20
χ²/ndf = 15.54
SNR = 1.08
r:1 c:2 m:0
Total events: 777610
Landau-Gauss fit, trigg.
L MPV = 70.82 ± 0.22
Max = 76.32 ± 0.23
χ²/ndf = 26.77
SNR = 0.90
r:1 c:3 m:0
Total events: 778805
Landau-Gauss fit, trigg.
L MPV = 65.97 ± 0.29
Max = 71.30 ± 0.31
χ²/ndf = 15.84
SNR = 0.90
r:0 c:0 m:0
Total events: 779149
Landau-Gauss fit, trigg.
L MPV = 68.24 ± 0.28
Max = 73.46 ± 0.30
χ²/ndf = 7.40
SNR = 1.13
r:0 c:1 m:0
Total events: 718952
Landau-Gauss fit, trigg.
L MPV = 66.20 ± 0.25
Max = 71.40 ± 0.27
χ²/ndf = 25.57
SNR = 1.80
r:0 c:2 m:0
Total events: 778065
Landau-Gauss fit, trigg.
L MPV = 70.33 ± 0.23
Max = 75.38 ± 0.25
χ²/ndf = 10.02
SNR = 1.23
r:0 c:3 m:0
Total events: 778854
Landau-Gauss fit, trigg.
L MPV = 72.73 ± 0.35
Max = 78.39 ± 0.38
χ²/ndf = 8.96
SNR = 1.26
counts
ADC (arb. units)

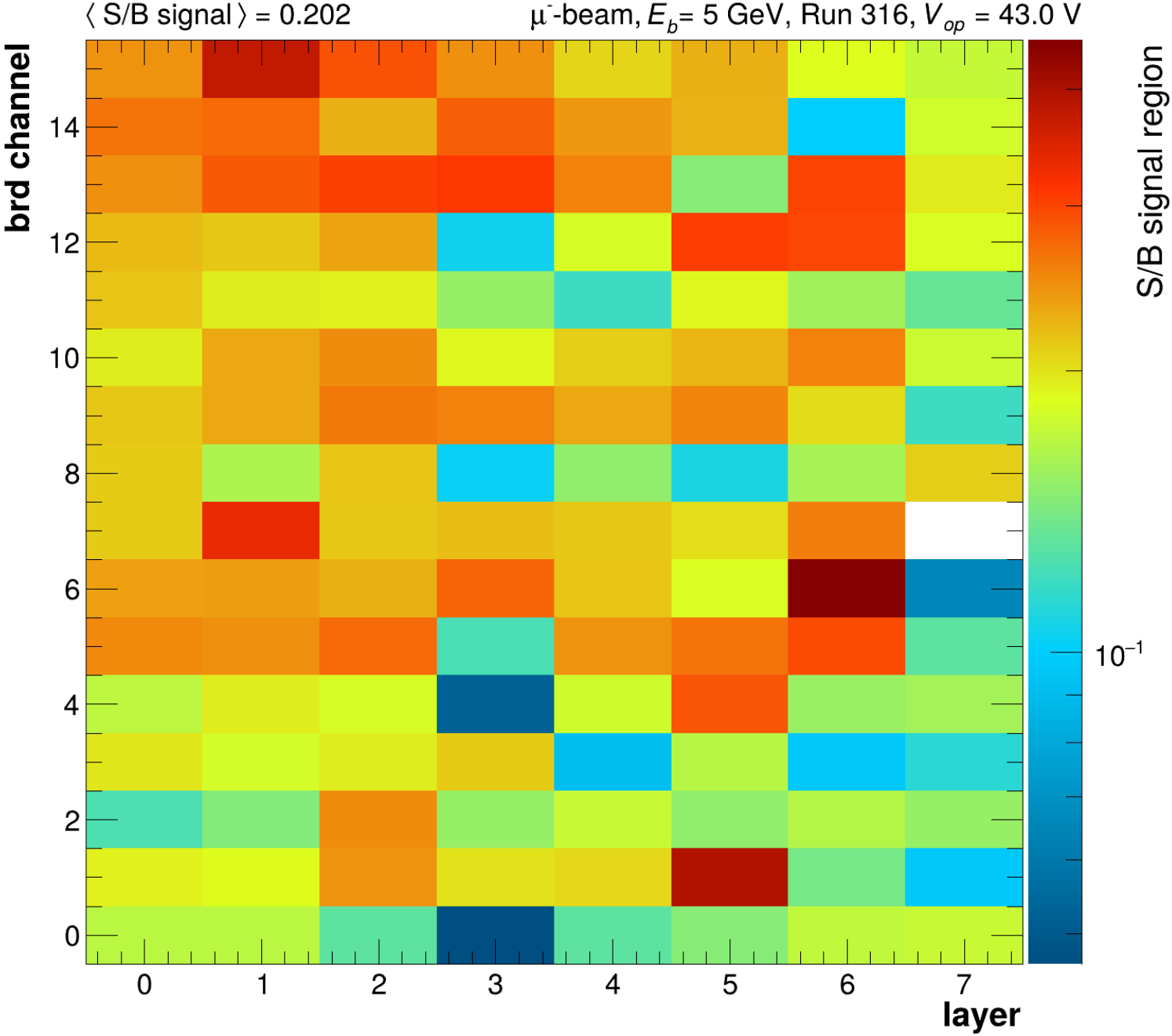

r:1 c:0 m:1
Total events: 551777
Landau-Gauss fit, trigg.
L MPV = 70.83 ± 0.23
Max = 70.42 ± 0.23
χ²/ndf = 16.57
SNR = 0.87
r:1 c:1 m:1
Total events: 543239
Landau-Gauss fit, trigg.
L MPV = 79.44 ± 0.26
Max = 85.39 ± 0.28
χ²/ndf = 15.72
SNR = 0.97
r:1 c:2 m:1
Total events: 538935
Landau-Gauss fit, trigg.
L MPV = 75.92 ± 0.24
Max = 81.44 ± 0.26
χ²/ndf = 17.47
SNR = 0.97
r:1 c:3 m:1 layer 7
Total events: 532434
Landau-Gauss fit, trigg.
L MPV = 74.13 ± 0.48
Max = 78.77 ± 0.51
χ²/ndf = 15.37
SNR = 0.94
μ⁻-beam, E_b = 5 GeV
Run 316, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 547706
Landau-Gauss fit, trigg.
L MPV = 64.09 ± 0.26
Max = 64.24 ± 0.26
χ²/ndf = 24.14
SNR = 0.74
r:0 c:1 m:1
Total events: 542878
Landau-Gauss fit, trigg.
L MPV = 80.04 ± 0.34
Max = 85.97 ± 0.37
χ²/ndf = 18.40
SNR = 2.11
r:0 c:2 m:1
Total events: 488256
Landau-Gauss fit, trigg.
L MPV = 79.77 ± 0.33
Max = 86.50 ± 0.36
χ²/ndf = 17.71
SNR = 1.09
r:0 c:3 m:1
Total events: 345060
Landau-Gauss fit, trigg.
L MPV = 84.78 ± 0.49
Max = 91.12 ± 0.53
χ²/ndf = 11.71
SNR = 0.85
r:1 c:0 m:0
Total events: 779177
Landau-Gauss fit, trigg.
L MPV = 75.26 ± 0.35
Max = 80.92 ± 0.38
χ²/ndf = 12.38
SNR = 0.80
r:1 c:1 m:0
Total events: 778011
Landau-Gauss fit, trigg.
L MPV = 77.10 ± 0.31
Max = 82.96 ± 0.33
χ²/ndf = 22.58
SNR = 1.14
r:1 c:2 m:0
Total events: 777832
Landau-Gauss fit, trigg.
L MPV = 80.11 ± 0.32
Max = 86.19 ± 0.34
χ²/ndf = 23.88
SNR = 4.25
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 779115
Landau-Gauss fit, trigg.
L MPV = 69.80 ± 0.40
Max = 74.95 ± 0.43
χ²/ndf = 9.60
SNR = 0.81
r:0 c:1 m:0
Total events: 778048
Landau-Gauss fit, trigg.
L MPV = 78.26 ± 0.37
Max = 83.91 ± 0.39
χ²/ndf = 13.38
SNR = 1.13
r:0 c:2 m:0
Total events: 758360
Landau-Gauss fit, trigg.
L MPV = 78.80 ± 0.37
Max = 85.16 ± 0.40
χ²/ndf = 12.88
SNR = 0.93
r:0 c:3 m:0
Total events: 687320
Landau-Gauss fit, trigg.
L MPV = 75.84 ± 0.41
Max = 81.63 ± 0.44
χ²/ndf = 8.85
SNR = 0.89
counts
ADC (arb. units)

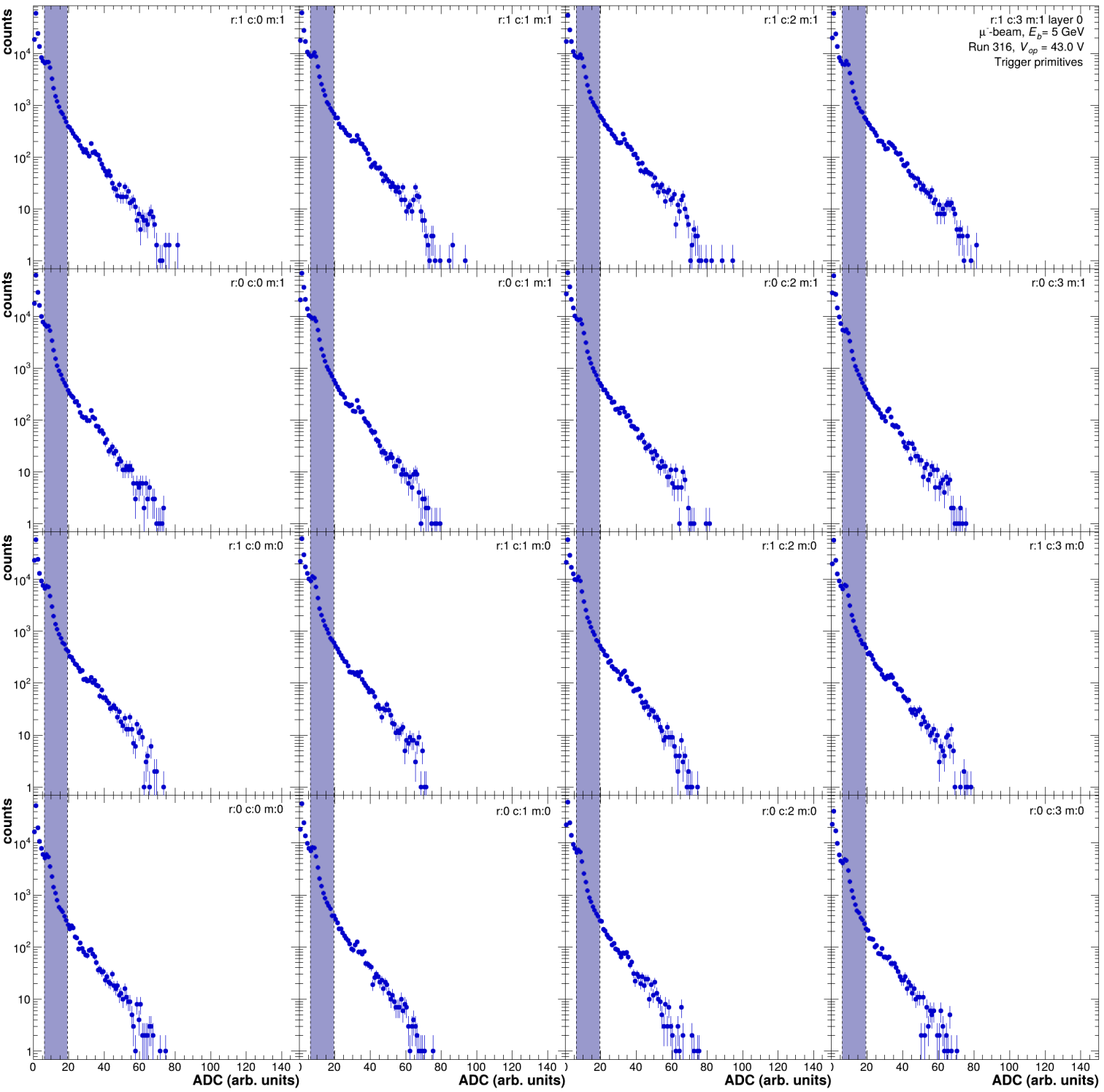

μ⁻-beam, E_b= 5 GeV, Run 316, V_{op} = 43.0 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
400
350
300
250
200

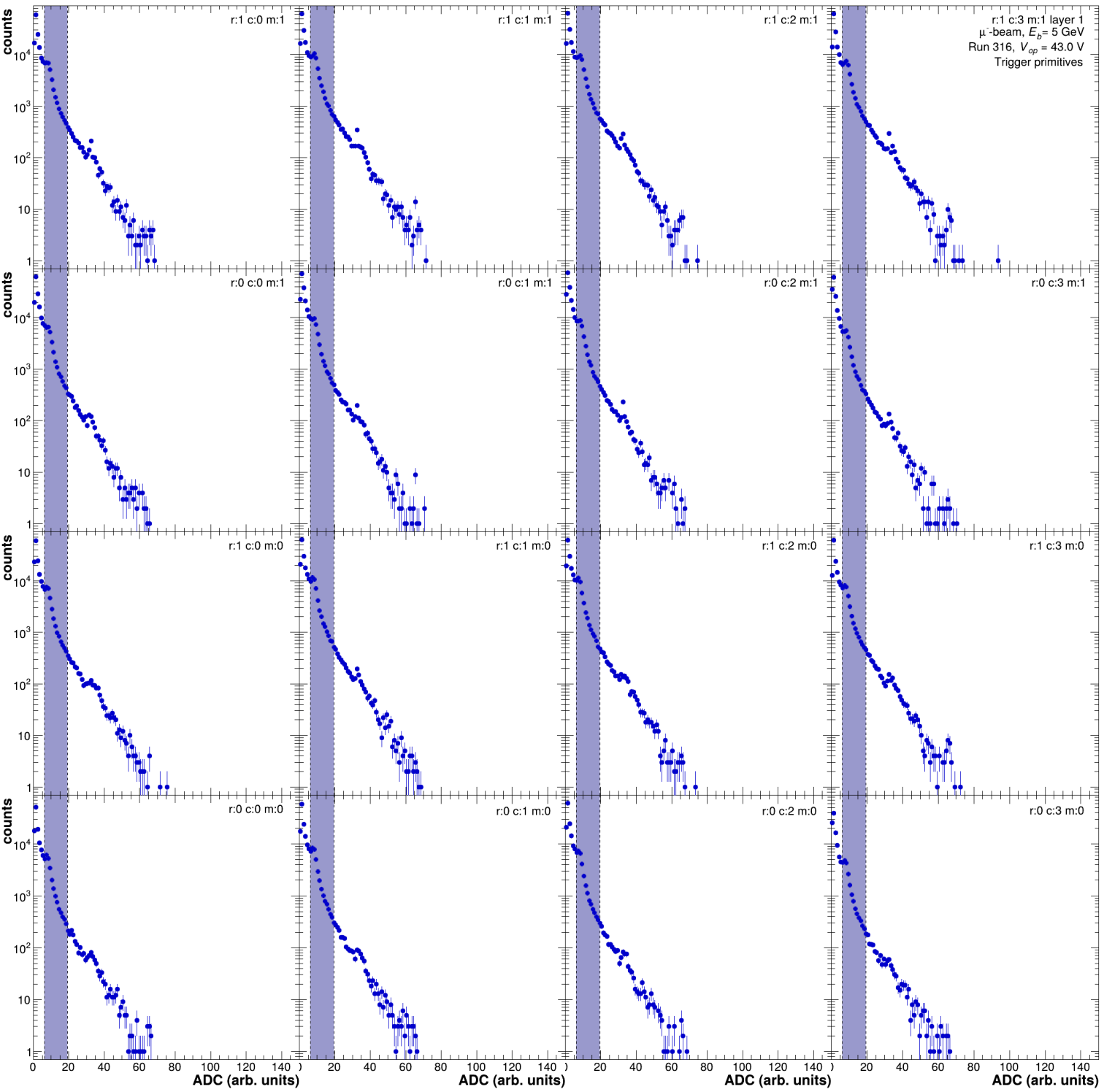

evt. collected = 779475
μ⁻-beam, E_b= 5 GeV, Run 316, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
counts

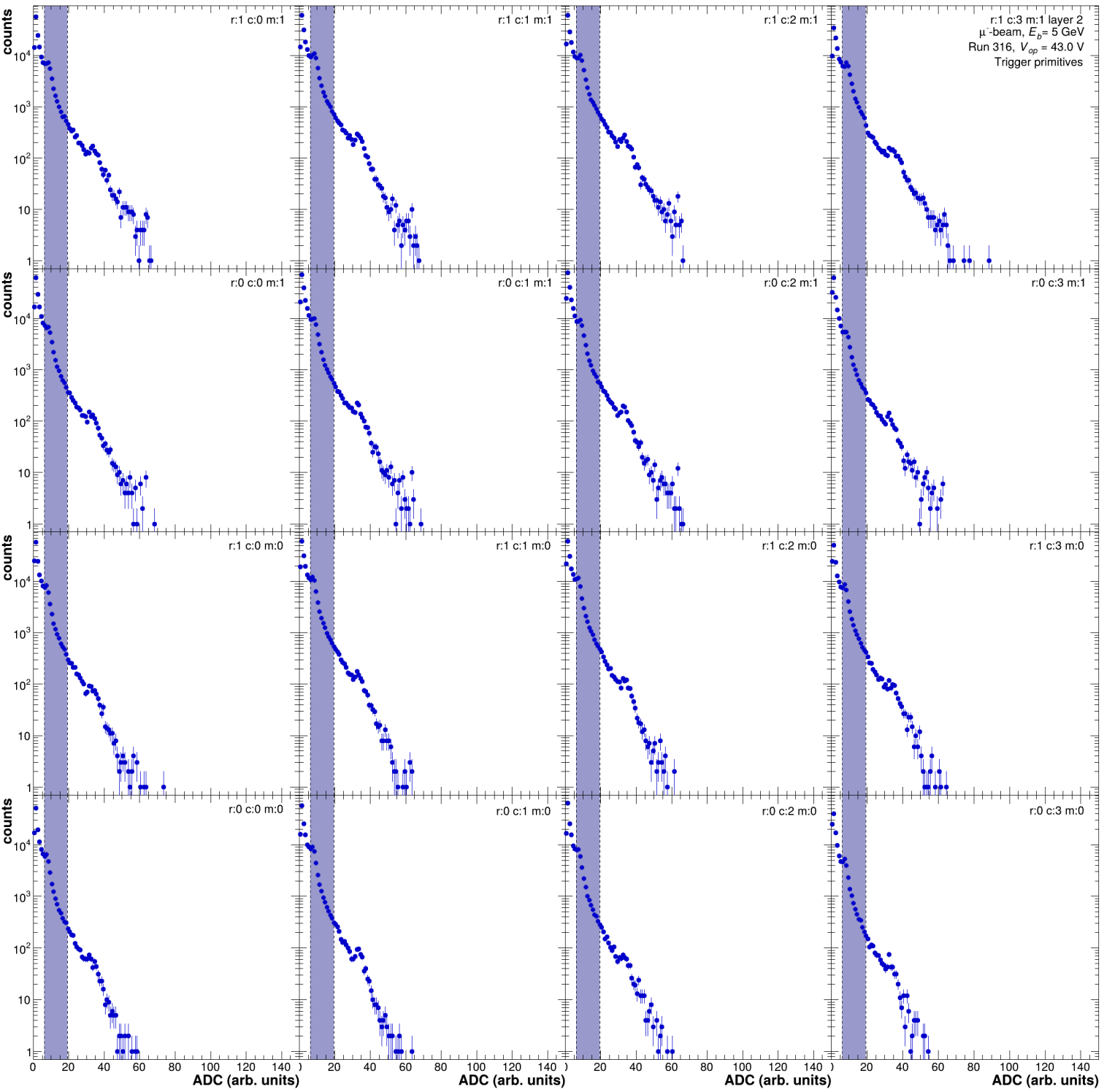

μ⁻-beam, E_b= 5 GeV, Run 316, V_{op} = 43.0 V
brd channel
layers
SNR
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
2
4
6
8
10

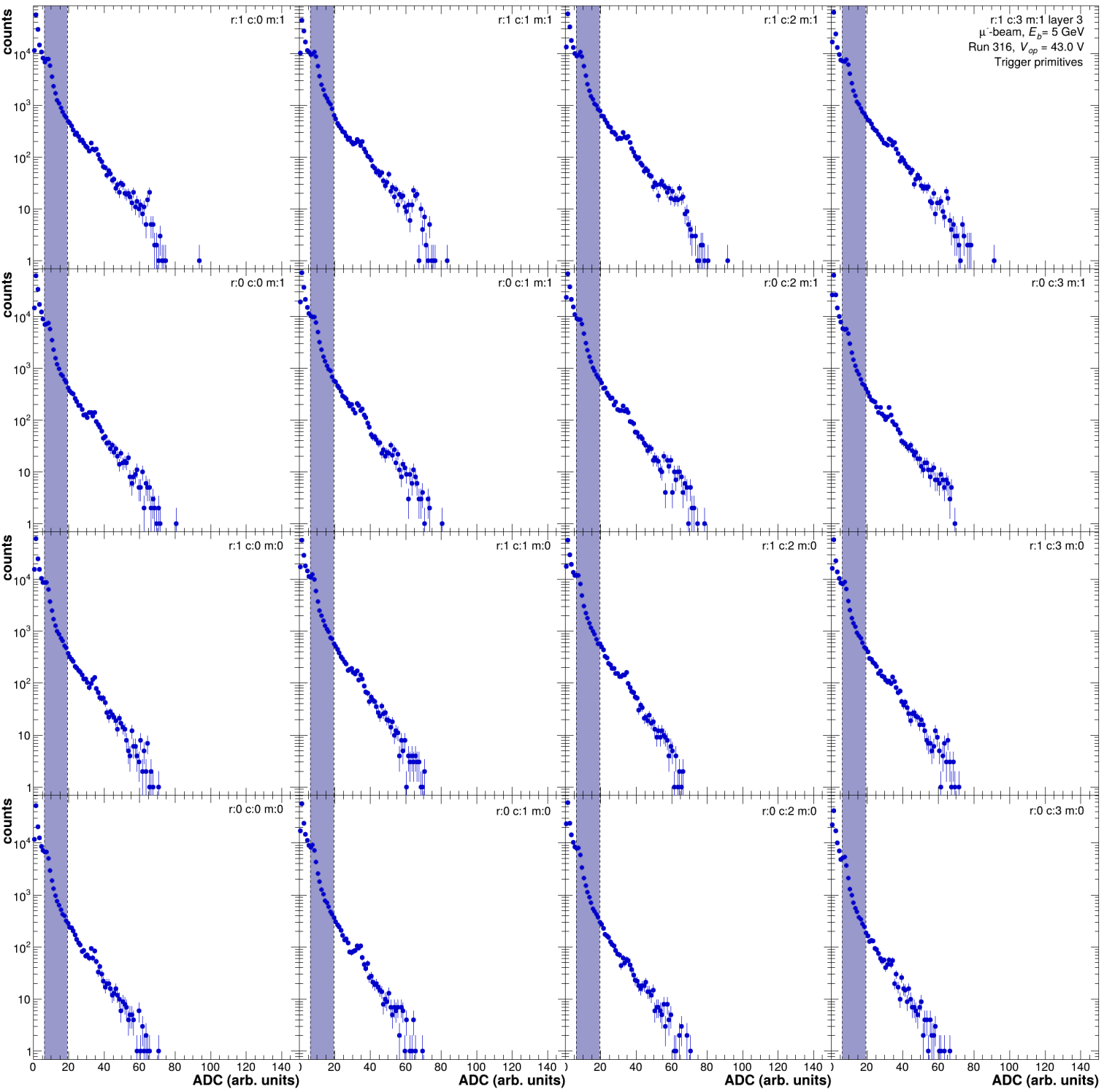

⟨ S/B noise ⟩ = 0.027
μ⁻-beam, E_b= 5 GeV, Run 316, V_{op} = 43.0 V
brd channel
layer
S/B noise region
10^{-1}
10^{-2}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

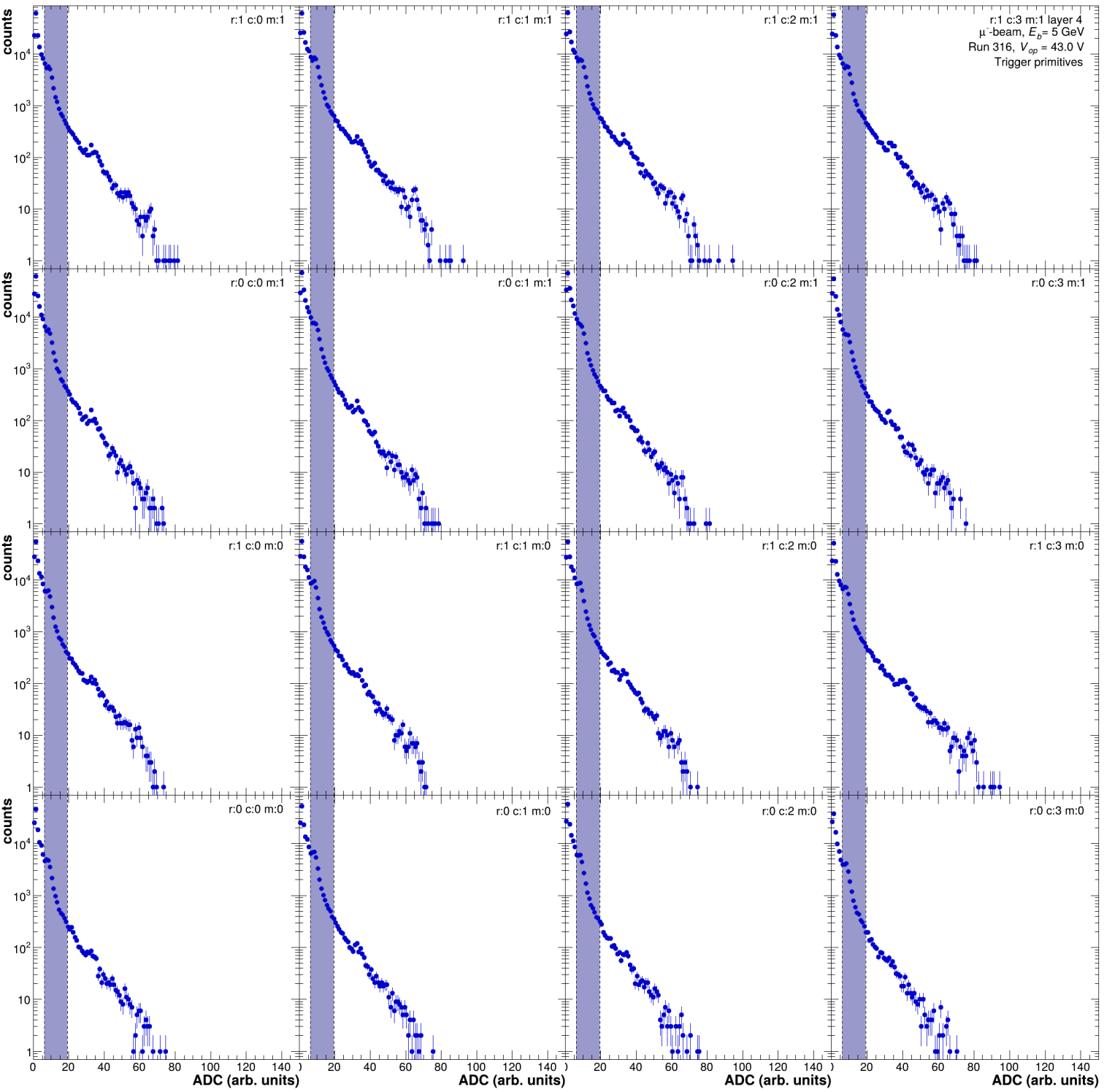

⟨ S/B signal ⟩ = 0.202
μ⁻-beam, E_b= 5 GeV, Run 316, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B signal region
10^{-1}

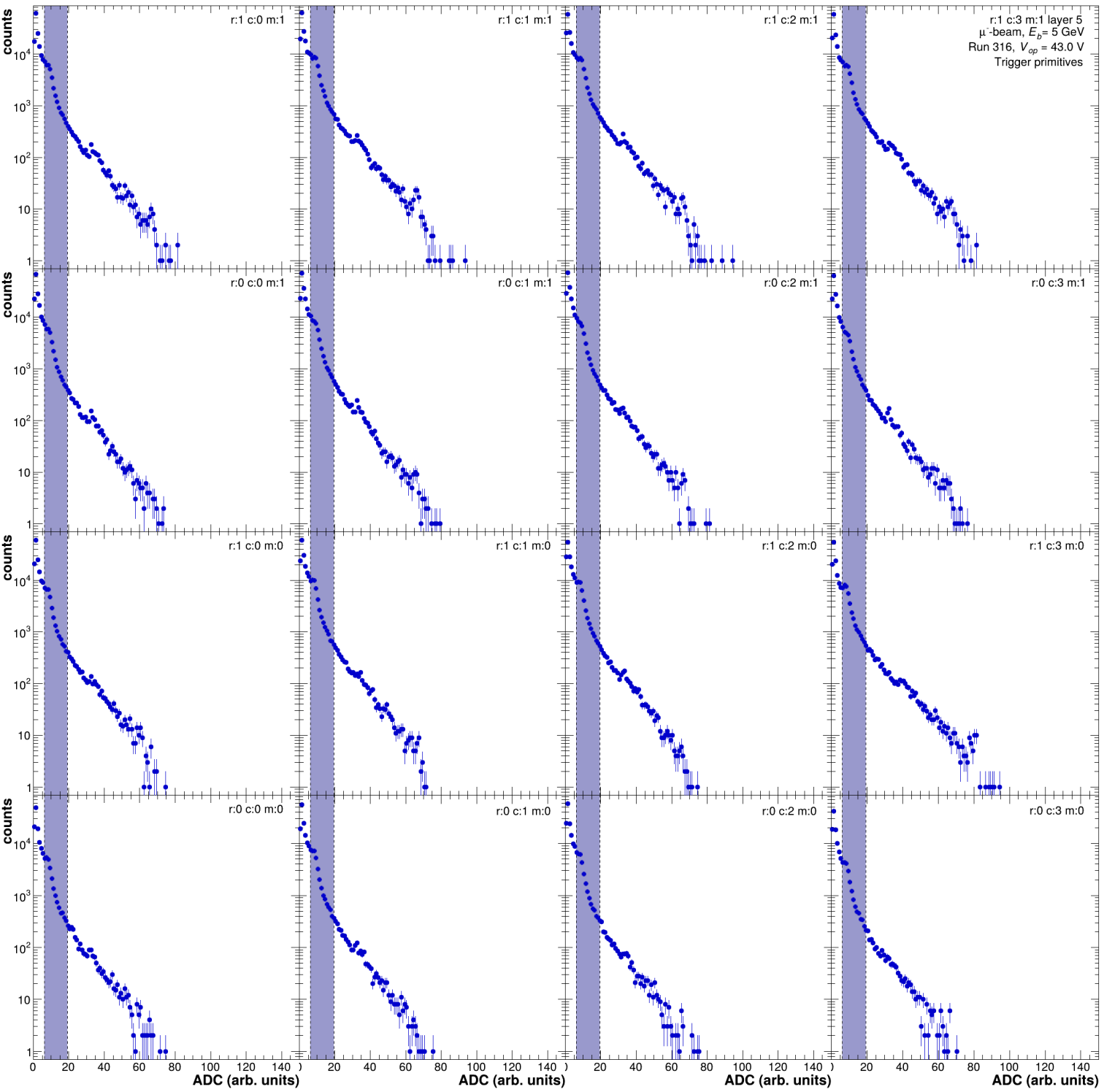

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 316, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

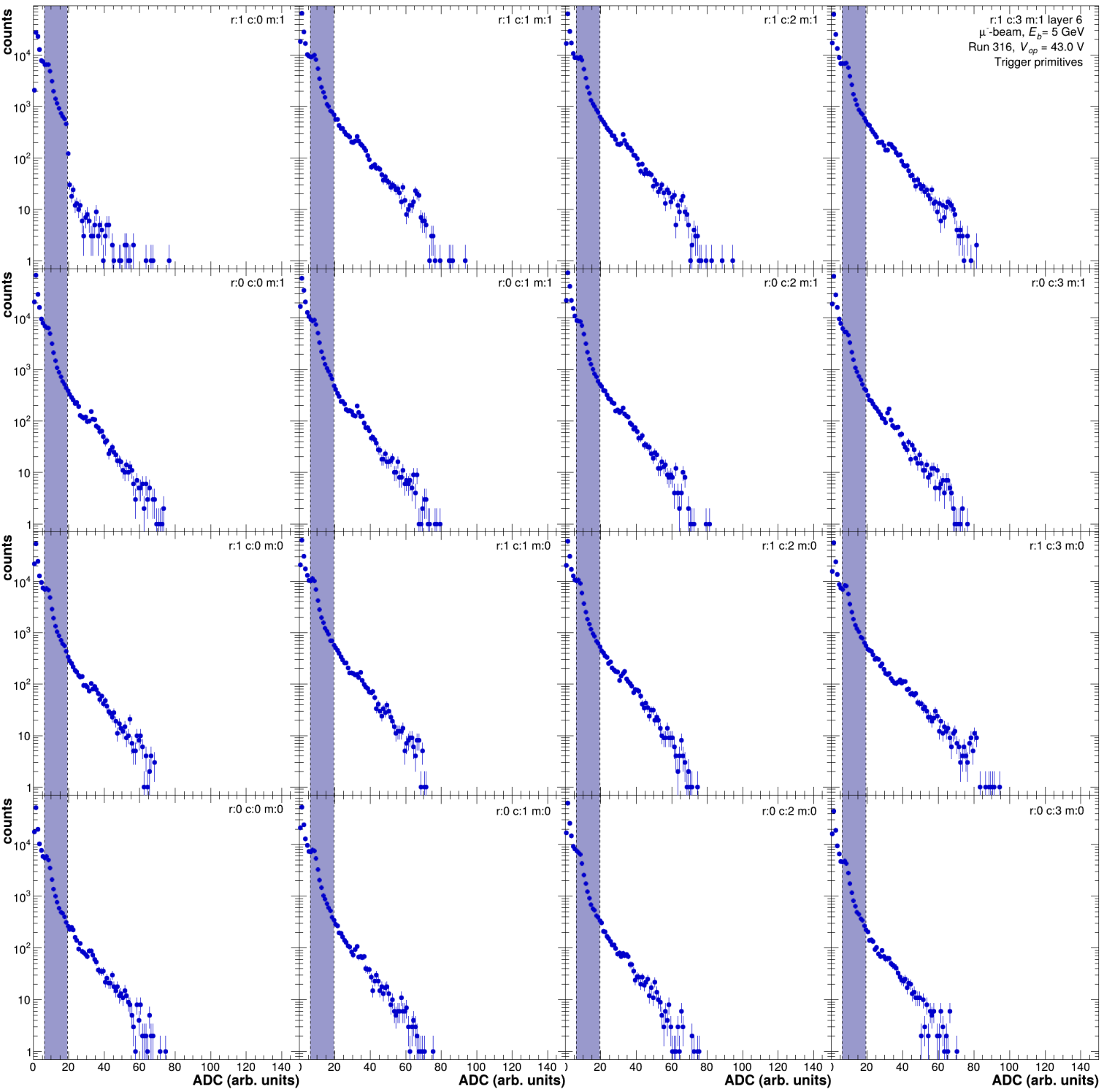

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 316, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

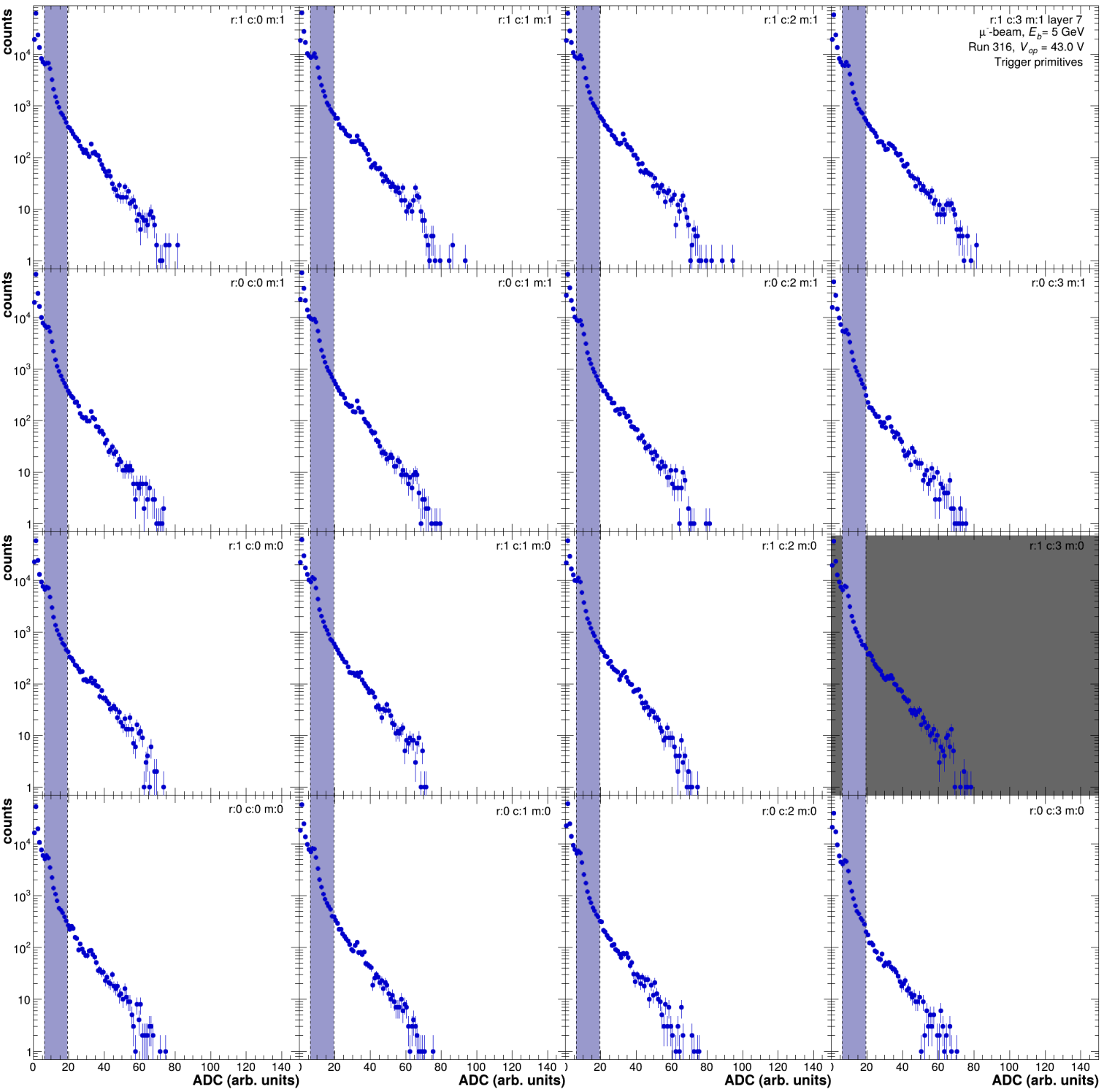

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 316, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 316, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 316, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 316, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 316, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 316, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)